

Closed Motion–Staticity Geometry: Null Boundaries and Boundary-to-Bulk Rigidity

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Abstract We develop a closed-boundary extension of the compact motion–staticity formulation of free relativistic dynamics. The starting point is the compact state

$$(D, S) \in \mathbb{R}^3 \oplus \mathbb{R}, \quad |D|^2 + S^2 = 1, \quad S > 0,$$

whose interior is projectively equivalent to the future unit timelike hyperboloid. We adjoin the boundary $S = 0$ and study which structures of the massive theory survive there, which information is lost, and under what additional response conditions the massive bulk can be reconstructed.

The first result is a boundary-underdetermination statement. For a general compact convex motion–staticity body K , whose free Hamiltonian is

$$H_K(p, Q) = c h_K(p, Q),$$

the $Q = 0$ dispersion depends only on the projection $\Omega = \text{proj}_D K$:

$$H_K(p, 0) = c h_\Omega(p).$$

Consequently, distinct massive compact-state geometries may share the same projective null boundary and the same limiting massless dispersion. The null boundary alone therefore does not determine the massive interior.

We then introduce the normalized energy–staticity reciprocity defect

$$\mathfrak{R} := hS - 1, \quad h := \frac{H}{cQ}.$$

In velocity variables it satisfies the exact identity

$$\mathfrak{R} = (D \cdot \nabla_D - 2) \frac{1 - S^2}{2}.$$

Hence exact reciprocity, $\mathfrak{R} = 0$, forces $1 - S^2$ to be homogeneous of degree two. Under C^2 regularity at the unique rest state, this implies

$$S^2 = 1 - D^T A D$$

for a positive-definite matrix A . Thus energy–staticity reciprocity selects an ellipsoidal compact state; ordinary spatial isotropy and calibration reduce this ellipsoid to the round motion–staticity hemisphere and recover the standard relativistic square-root Hamiltonian.

A further extension is obtained by imposing the established hyperbolic affine-sphere volume condition on the Legendre potential $u = -S$. In that class, the reciprocity defect is related directly to the Pick cubic tensor, so that derivatives of a scalar energy–staticity response reconstruct the local nonquadratic anisotropy of the observer geometry. Under smooth strictly convex boundary regularity, the renormalized boundary quantity

$$\beta(\xi) := \lim_{D \rightarrow \xi \in \partial\Omega} (h(D)S(D) - 1)$$

is determined by the centro-affine curvature $\mathcal{K}_0$ of $\partial\Omega$ through

$$1 + \beta = \rho^{2/(n+1)} \mathcal{K}_0^{-1/(n+1)},$$

with ρ the affine-volume normalization. Near the round state, the linearized boundary response obeys

$$\delta\beta = \frac{1}{n+1} (\Delta_{S^{n-1}} + 2n) \delta s,$$

where s is the boundary support function. Its kernel consists precisely of the infinitesimal ellipsoidal modes.

The resulting picture separates boundary universality from bulk rigidity. Projective null geometry is shared by a broad class of compact-state models, whereas suitably normalized energy–staticity and support-response data recover the quadric Lorentzian case. The construction is intended as a structural boundary-to-bulk extension of motion–staticity geometry, not as a modified-relativity phenomenology.

Keywords: motion–staticity geometry; compact-state dynamics; null boundary; Hamiltonian rigidity; convex duality; energy–staticity reciprocity; affine sphere; centro-affine geometry; boundary-to-bulk reconstruction; Hilbert geometry.

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1 Introduction

Relativistic velocity space admits several mathematically equivalent descriptions. In the standard formulation, massive observer states form the future unit timelike hyperboloid, while massless directions are represented by the projectivized future null cone. Hyperbolic velocity geometry, rapidity, Lorentz transformations, and the causal light cone are therefore normally introduced within an ambient Lorentzian structure. An alternative logical organization was developed in the compact motion–staticity formulation of free massive dynamics [?]. There the primitive state is a bounded vector–scalar pair

$$(D, S) \in \mathbb{R}^3 \oplus \mathbb{R}, \quad |D|^2 + S^2 = 1, \quad S > 0,$$

with operational calibration

$$D = \frac{v}{c}, \quad S = \frac{d\tau}{dt}.$$

The same state admits the projective representation

$$(D, S) \mapsto \frac{1}{S}(1, D),$$

which identifies the open hemisphere with the standard future unit hyperboloid. The compact variables therefore do not modify the free massive phenomenology of special relativity. Their role is instead reconstructive: the vector and scalar entries are required to arise consistently from Hamiltonian velocity and the Legendre remainder, and this compatibility imposes rigidity on the free-particle dispersion relation.

In the Euclidean compact state, Hamiltonian consistency takes the form

$$\frac{|\nabla_p H|^2}{c^2} + \frac{(H - p \cdot \nabla_p H)^2}{c^2 Q^2} = 1.$$

Under the regularity and nondegeneracy assumptions stated in [?], the nontrivial global branch is

$$H(p) = c \sqrt{|p|^2 + Q^2}.$$

The same work also placed this result in a broader convex framework. If $K \subset \mathbb{R}^3 \oplus \mathbb{R}$ is a compact convex motion–staticity body and h_K denotes its support function, then the associated free Hamiltonian is

$$H_K(p, Q) = c h_K(p, Q).$$

This extension is important conceptually because it separates the general convex duality principle from the specifically Euclidean choice of compact state. Compactness alone does not select the relativistic square root; the geometry of the compact body matters.

The present paper addresses a question that was deliberately left open in the massive theory: what is the status of the boundary $S = 0$? For fixed $Q > 0$, the limit $S \rightarrow 0^+$ is mapped projectively to infinite timelike rapidity and should not be identified without further analysis with a finite-energy massless particle. Nevertheless, the closed compact state

$$\overline{\mathcal{S}}_+ = \{(D, S) : |D|^2 + S^2 = 1, S \geq 0\}$$

contains a distinguished boundary

$$\partial \overline{\mathcal{S}}_+ = \{(D, 0) : |D| = 1\} \simeq S^1,$$

and the projective image of this boundary is the space of future null directions. It is therefore natural to ask whether the closed state carries more information than a convenient compactification of the massive hyperboloid.

There are two logically distinct questions. The first is a boundary-limit problem: which features of the massive Hamiltonian and observer geometry have well-defined descendants at $S = 0$? The second is an inverse problem: to what extent can the massive interior be reconstructed from null-boundary information? These questions should not be conflated. A projective boundary may be universal across a broad class of massive theories even when the corresponding bulk dynamics is not.

A first result of this paper makes this distinction precise. Let

$$\Omega := \text{proj}_D K$$

be the projection of a compact convex state body onto its motion sector. Since

$$H_K(p, Q) = c h_K(p, Q),$$

setting $Q = 0$ gives

$$H_K(p, 0) = c h_\Omega(p).$$

Thus the limiting homogeneous dispersion depends only on Ω , not on the full vertical profile of K . Distinct compact-state bodies with the same motion projection can therefore share the same null-boundary dispersion. This establishes a basic boundary-underdetermination principle: projective null geometry alone does not reconstruct massive dynamics.

The missing information is supplied by the staticity response. Introduce dimensionless momentum and energy,

$$k := \frac{p}{Q}, \quad h(k) := \frac{H(p)}{cQ}.$$

On the regular Legendre branch,

$$D = \nabla_k h, \quad S = h - k \cdot \nabla_k h.$$

We define the energy–staticity reciprocity defect by

$$\mathfrak{R} := hS - 1.$$

In the standard relativistic solution, $h = 1/S$ and $\mathfrak{R}$ vanishes identically. The significance of this relation is not restricted to the relativistic solution. In velocity variables one obtains the exact identity

$$\mathfrak{R} = (D \cdot \nabla_D - 2) \frac{1 - S^2}{2}.$$

Consequently,

$$\mathfrak{R} = 0$$

is precisely an Euler homogeneity equation: $1 - S^2$ must be homogeneous of degree two. If the compact state possesses a single regular rest point and S is C^2 there, degree-two homogeneity forces

$$1 - S^2 = D^T A D$$

with A positive definite. Exact energy–staticity reciprocity therefore selects an ellipsoid. Ordinary spatial isotropy and the velocity calibration then give $A = I$, recovering

$$|D|^2 + S^2 = 1.$$

This provides a reverse characterization of the Euclidean motion–staticity state: the round hemisphere need not be taken as the final geometric input if one instead assumes a scalar reciprocity principle together with regularity at rest.

This rigidity also clarifies the role of the rest state. Away from $D = 0$, degree-two homogeneous functions may retain genuine directional dependence and define Finsler-type structures on a slit domain. Smooth collapse of all such directions to a single C^2 rest state eliminates this freedom and leaves a quadratic form. The rest state is therefore not merely the point of zero velocity; it is the point at which directional kinematic data must admit a common regular extension.

A second part of the paper investigates a broader completion in which the Legendre potential

$$u(D) := -S(D)$$

is required to define a hyperbolic affine sphere. This is not a new construction of affine-sphere relativity. Hyperbolic affine spheres, their Monge–Ampère equation, and their role in Lorentz–Finsler observer geometry have been developed independently in affine differential geometry and in the work of Minguzzi and others [?, ?]. The purpose here is instead to determine what the motion–staticity reciprocity scalar measures inside that established class.

With affine-volume normalization ρ , the potential satisfies

$$\det D^2 u = \rho^2 (-u)^{-(n+2)}.$$

The associated Pick cubic tensor measures the departure of the affine sphere from a hyperquadric. We show that the energy–staticity reciprocity defect is directly related to this cubic anisotropy. In a standard Pick normalization,

$$\partial_i \partial_j \mathfrak{R} = 2S^2 D^k c_{ijk}.$$

At the rest state this reduces to a local cubic tomography formula: the third derivatives of the scalar response $\mathfrak{R}$ reconstruct the Pick cubic. Thus a tensorial departure from quadratic observer geometry can be detected by derivatives of a single scalar energy–staticity quantity.

The same structure has a boundary counterpart. Suppose that Ω has smooth strictly convex boundary and let

$$s(\nu) = \sup_{D \in \Omega} D \cdot \nu$$

be its support function. Under the boundary regularity required for the affine-sphere asymptotics, define the renormalized boundary reciprocity response

$$\beta(\xi) = \lim_{D \rightarrow \xi \in \partial\Omega} \mathfrak{R}(D).$$

Its leading boundary behavior is controlled by the centro-affine curvature

$$\mathcal{K}_0 = \frac{\kappa}{s^{n+1}},$$

where κ is the Euclidean Gauss curvature of $\partial\Omega$. More precisely,

$$1 + \beta = \rho^{2/(n+1)} \mathcal{K}_0^{-1/(n+1)}.$$

Thus β is not an arbitrary additional boundary field. It is a response-theoretic representation of a classical centro-affine boundary invariant. Constant β corresponds to constant centro-affine curvature and hence, under the standard convexity hypotheses, to a centered ellipsoid.

Near the round state this relation becomes a linear boundary tomography problem. Writing

$$s = 1 + \varepsilon f,$$

one obtains

$$\delta\beta = \frac{1}{n+1} (\Delta_{S^{n-1}} + 2n) f.$$

The kernel consists of the degree-two spherical harmonics, exactly the infinitesimal ellipsoidal modes generated by affine recalibration. All genuinely nonellipsoidal modes are therefore visible to the linearized boundary response. In three spatial dimensions,

$$\delta\beta = \frac{1}{4} (\Delta_{S^2} + 6) f.$$

The resulting hierarchy separates three distinct levels of boundary information. The topology of the null boundary is highly universal. Its support function determines the projective convex boundary and the limiting homogeneous dispersion, but not the full massive staticity profile. Energy–staticity reciprocity adds radial information and yields a quadric rigidity condition. In the affine-sphere completion, the renormalized boundary reciprocity response further becomes a centro-affine curvature datum and leads naturally to a nonlinear boundary-reconstruction problem of Minkowski type.

The paper is organized as follows. Section 2 introduces the closed compact-state geometry and its projective null boundary. Section 3 studies the dynamical degeneration at zero staticity. Section 4 establishes boundary underdetermination for general compact convex states. Section 5 develops energy–staticity reciprocity and proves the associated ellipsoidal rigidity theorem. Section 6 introduces the affine-sphere completion and relates the reciprocity defect to the Pick cubic. Section 7 derives the renormalized boundary response and its centro-affine interpretation. Section 8 studies the near-ellipsoidal boundary tomography problem. Section 9 summarizes the boundary-to-bulk picture, its assumptions, and its limitations.

Throughout the paper, the boundary $S = 0$ is treated as a projective null boundary of the fixed- Q massive state space. No claim is made that adjoining this boundary alone constructs a complete finite-energy massless particle theory. Likewise, the affine-sphere condition is an additional geometric completion, not a consequence of compact-state duality by itself. The purpose is narrower: to determine precisely which boundary structures are universal, which massive data they fail to retain, and which additional response relations recover the quadric Lorentzian case.

2 Closed Compact-State Geometry and the Projective Null Boundary

The compact motion–staticity state introduced in the massive theory is the open upper hemisphere

$$\mathcal{S}_+ = \{(D, S) \in \mathbb{R}^3 \oplus \mathbb{R} : |D|^2 + S^2 = 1, \quad S > 0\}.$$

The restriction $S > 0$ is appropriate for the free massive sector, since under the operational calibration

$$D = \frac{v}{c}, \quad S = \frac{d\tau}{dt},$$

one has $|D| < 1$ and $S > 0$. The present paper enlarges this state space only as a geometric compactification. We define its closure by

$$\overline{\mathcal{S}}_+ = \{(D, S) \in \mathbb{R}^3 \oplus \mathbb{R} : |D|^2 + S^2 = 1, \quad S \geq 0\}.$$

Its boundary is

$$\partial \overline{\mathcal{S}}_+ = \{(n, 0) : n \in S^2\} \simeq S^2.$$

The purpose of this section is to identify the precise geometric meaning of this boundary. The main point is that the interior remains the usual massive hyperbolic observer space, whereas the boundary parametrizes projective future null directions. The staticity variable S is simultaneously the scalar compact-state coordinate and a defining function for this conformal boundary.

2.1 Closed motion–staticity states

Let

$$B^3 = \{D \in \mathbb{R}^3 : |D| < 1\}.$$

On the upper hemisphere the scalar coordinate is determined by the motion coordinate:

$$S(D) = \sqrt{1 - |D|^2}.$$

Hence the projection

$$\pi : \mathcal{S}_+ \longrightarrow B^3, \quad \pi(D, S) = D,$$

is a diffeomorphism, and the closed state space is naturally identified with the closed unit ball

$$\overline{B}^3 = \{D \in \mathbb{R}^3 : |D| \leq 1\}.$$

The interior point

$$(D, S) = (0, 1)$$

is the unique rest state. The opposite limiting locus,

$$S = 0, \quad |D| = 1,$$

is not a single state but a two-sphere of limiting motion directions.

It is useful to introduce the radial motion variable

$$r := |D| = \sqrt{1 - S^2}.$$

The two distinguished ends of the closed state geometry are then

$$r = 0, \quad S = 1,$$

and

$$r = 1, \quad S = 0.$$

The first is an interior point at which the motion direction is undefined, whereas the second is a genuine boundary on which only the direction survives.

A radial rapidity coordinate $\chi \geq 0$ may be introduced in the interior by

$$r = \tanh \chi.$$

The compact-state constraint then gives

$$S = \sqrt{1 - \tanh^2 \chi} = \operatorname{sech} \chi.$$

Thus

$$\chi \rightarrow 0 \quad \Longleftrightarrow \quad (D, S) \rightarrow (0, 1),$$

while

$$\chi \rightarrow +\infty \quad \Longleftrightarrow \quad S \rightarrow 0^+.$$

Equivalently,

$$\cosh \chi = \frac{1}{S}, \quad \sinh \chi = \frac{|D|}{S}.$$

The compact staticity coordinate therefore exponentially resolves the infinite-rapidity end of the usual massive observer space.

The intrinsic round metric inherited from the Euclidean metric on $\mathbb{R}^3 \oplus \mathbb{R}$ is

$$\bar{h} = |dD|^2 + dS^2$$

restricted to

$$|D|^2 + S^2 = 1.$$

Since

$$dS = -\frac{D \cdot dD}{S},$$

one may write

$$\bar{h} = |dD|^2 + \frac{(D \cdot dD)^2}{S^2}.$$

In spherical variables,

$$D = \sin \theta n, \quad S = \cos \theta, \quad 0 \leq \theta \leq \frac{\pi}{2},$$

this becomes the standard round metric

$$\bar{h} = d\theta^2 + \sin^2 \theta d\Omega_2^2.$$

The boundary $S = 0$ is therefore the equatorial two-sphere of the closed hemisphere.

This compact metric should not be confused with the Lorentzian spacetime metric or with the hyperbolic metric on massive observer space. It is a metric on the compact state manifold itself. The relation between the compact and hyperbolic geometries is instead projective and conformal, as described next.

2.2 Projective compactification and the null boundary

Let $\mathbb{R}^{1,3}$ carry the Minkowski quadratic form

$$\eta(U, U) = (U^0)^2 - |\mathbf{U}|^2.$$

The future unit timelike hyperboloid is

$$\mathbb{H}_+^3 = \{U = (U^0, \mathbf{U}) : \eta(U, U) = 1, \quad U^0 > 0\}.$$

Define

$$\Phi : \mathcal{S}_+ \longrightarrow \mathbb{H}_+^3$$

by

$$\Phi(D, S) = \frac{1}{S}(1, D).$$

Then

$$\eta(\Phi, \Phi) = \frac{1 - |D|^2}{S^2} = 1,$$

and the time component is positive. Conversely, if $U \in \mathbb{H}_+^3$, then

$$S = \frac{1}{U^0}, \quad D = \frac{\mathbf{U}}{U^0}$$

satisfy

$$|D|^2 + S^2 = 1.$$

Thus Φ is a diffeomorphism between the open compact state and the usual future unit hyperboloid.

Proposition 2.1 (Projective hyperbolic compactification). *The map*

$$\Phi(D, S) = S^{-1}(1, D)$$

identifies the interior $\mathcal{S}_+$ with the future unit timelike hyperboloid. Under this identification, the boundary $S = 0$ corresponds to the projectivized future null cone.

Proof. The interior statement follows from the preceding calculation. Let

$$(D_j, S_j) \in \mathcal{S}_+$$

converge to a boundary point

$$(n, 0), \quad |n| = 1.$$

Although

$$\Phi(D_j, S_j) = \frac{1}{S_j}(1, D_j)$$

diverges in $\mathbb{R}^{1,3}$, its projective class is unchanged by multiplication by S_j . Hence

$$[\Phi(D_j, S_j)] = [(1, D_j)] \longrightarrow [(1, n)].$$

The limiting vector

$$N(n) := (1, n)$$

is future null because

$$\eta(N(n), N(n)) = 1 - |n|^2 = 0.$$

Conversely, every future null ray possesses a unique representative of the form $(1, n)$ with $|n| = 1$. Therefore

$$\partial \bar{\mathcal{S}}_+ \simeq \mathbb{P}(\mathcal{N}^+),$$

where $\mathcal{N}^+$ denotes the future null cone. □

The proposition makes clear why the boundary should be regarded as a projective null boundary rather than as an additional ordinary massive shell. Along a fixed- $Q > 0$ massive branch,

$$U^0 = \frac{1}{S}$$

diverges when $S \rightarrow 0^+$. The finite boundary object is not the four-velocity itself but its ray. In particular, the operation

$$S \rightarrow 0$$

at fixed positive intrinsic scale Q should not by itself be identified with the construction of a finite-energy massless particle. What survives canonically is the null direction.

The hyperbolic metric induced on $\mathbb{H}_+^3$ pulls back to a conformal rescaling of the compact round metric. In rapidity coordinates,

$$h_{\text{hyp}} = d\chi^2 + \sinh^2 \chi d\Omega_2^2.$$

Using

$$S = \text{sech } \chi, \quad \sin \theta = \tanh \chi,$$

one finds

$$d\theta = S d\chi$$

and hence

$$h_{\text{hyp}} = S^{-2} (d\theta^2 + \sin^2 \theta d\Omega_2^2).$$

Therefore

$$h_{\text{hyp}} = S^{-2} \bar{h}.$$

Thus the compact round metric is a conformal compactification of the standard hyperbolic observer metric.

Equivalently, in ball coordinates D ,

$$h_{\text{hyp}} = \frac{(1 - |D|^2)|dD|^2 + (D \cdot dD)^2}{(1 - |D|^2)^2}.$$

Since

$$S^2 = 1 - |D|^2,$$

the same expression may be written as

$$h_{\text{hyp}} = \frac{|dD|^2}{S^2} + \frac{(D \cdot dD)^2}{S^4}.$$

The metric diverges in ordinary compact coordinates as $S \rightarrow 0$, while

$$S^2 h_{\text{hyp}} = \bar{h}$$

extends smoothly and nondegenerately to the boundary. In this precise sense, S is a defining function for the conformal boundary of massive observer space.

Proposition 2.2 (Staticity as a conformal defining function). *On the closed motion–staticity hemisphere,*

$$S > 0 \quad \text{in the interior}, \quad S = 0 \quad \text{on the boundary},$$

and

$$dS \neq 0$$

in the normal direction to the boundary. Moreover,

$$\bar{h} = S^2 h_{\text{hyp}}$$

extends smoothly to $\partial \bar{S}_+$. Consequently, S is a conformal boundary-defining function for the hyperbolic massive observer geometry.

The conformal boundary inherits the round metric

$$d\Omega_2^2.$$

Under the projective identification above, it is the celestial sphere of future null directions. The same two-sphere can therefore be described in three equivalent ways:

$$\partial \bar{S}_+ = \{(n, 0) : |n| = 1\},$$

$$\partial \bar{S}_+ \simeq S^2,$$

and

$$\partial \bar{S}_+ \simeq \mathbb{P}(\mathcal{N}^+).$$

The role of $S = 0$ is consequently more precise than the statement that $|D|$ reaches unity. It is the locus at which the timelike projective representative escapes to infinite rapidity while its projective class converges to a finite null direction. The compact closure therefore separates the divergent scale of the massive four-velocity from the finite directional data retained at the boundary.

This distinction will be essential below. Section 3 studies which dynamical quantities possess a corresponding zero-staticity limit, while Section 4 shows that the projective boundary by itself does not contain enough information to reconstruct a general massive compact-state interior.

3 Zero-Staticity Dynamics

The geometric compactification of Section 2 identifies $S = 0$ with a projective null boundary, but geometry alone does not determine the dynamical law carried by that boundary. In the massive theory the scalar compact-state coordinate is represented canonically by

$$S_H(p) = \frac{H(p) - p \cdot \nabla_p H(p)}{cQ},$$

where $Q > 0$ is the intrinsic scale of the massive system. Accordingly, the unnormalized scalar Legendre remainder is

$$q_H(p) := \frac{H(p) - p \cdot \nabla_p H(p)}{c}.$$

The quantity q_H remains meaningful independently of the normalization by Q and is therefore the appropriate object for formulating the zero-staticity condition.

For the Euclidean compact state, massive representation consistency is

$$\frac{|\nabla_p H|^2}{c^2} + \frac{q_H^2}{Q^2} = 1.$$

On a sequence of massive states approaching the compact boundary, $S_H \rightarrow 0$ and hence $q_H \rightarrow 0$ at fixed Q . The corresponding limiting conditions are therefore

$$|\nabla_p H| = c, \quad H - p \cdot \nabla_p H = 0.$$

The second equation is the Euler equation for a positively homogeneous Hamiltonian of degree one. The first states that the Hamiltonian velocity has the universal limiting magnitude c .

These equations will be referred to as the zero-staticity Hamiltonian equations. They should not be interpreted as the result of substituting $Q = 0$ directly into the normalized massive equation, which is singular in that form. Rather, they isolate the finite relations that remain after the scalar Legendre coordinate has vanished.

3.1 Homogeneity at the boundary

Let

$$H_0 : \mathbb{R}^3 \setminus \{0\} \longrightarrow (0, \infty)$$

be a C^2 Hamiltonian intended to describe an intrinsic zero-staticity branch. We impose

$$H_0 - p \cdot \nabla H_0 = 0$$

and

$$|\nabla H_0| = c.$$

The first equation implies, by Euler's homogeneous-function theorem, that

$$H_0(\lambda p) = \lambda H_0(p), \quad \lambda > 0.$$

Thus the radial momentum scale is no longer an independent variable of the directional Hamiltonian geometry.

Writing

$$p = rn, \quad r = |p| > 0, \quad n \in S^2,$$

every positive degree-one homogeneous Hamiltonian has the form

$$H_0(p) = cr f(n)$$

for a positive function

$$f : S^2 \longrightarrow (0, \infty).$$

Its Euclidean momentum gradient decomposes into radial and angular parts:

$$\nabla_p H_0 = c(f(n)n + \nabla_{S^2} f(n)).$$

Consequently,

$$|\nabla_p H_0|^2 = c^2(f^2 + |\nabla_{S^2} f|^2).$$

The zero-staticity speed condition therefore reduces to

$$f^2 + |\nabla_{S^2} f|^2 = 1.$$

Theorem 3.1 (Positive zero-staticity rigidity). *Let*

$$H_0 \in C^2(\mathbb{R}^3 \setminus \{0\})$$

be positive and satisfy

$$H_0 - p \cdot \nabla H_0 = 0, \quad |\nabla H_0| = c.$$

Then

$$H_0(p) = c|p|.$$

Proof. By degree-one homogeneity,

$$H_0(p) = crf(n)$$

with $f > 0$ on S^2 . The speed condition gives

$$f^2 + |\nabla_{S^2} f|^2 = 1,$$

and hence

$$0 < f \leq 1.$$

Since S^2 is compact, f attains a minimum at some $n_0 \in S^2$. At that point,

$$\nabla_{S^2} f(n_0) = 0.$$

The preceding equation therefore gives

$$f(n_0)^2 = 1.$$

Positivity implies

$$f(n_0) = 1.$$

Since $f(n) \leq 1$ everywhere while its global minimum is already equal to 1, one must have

$$f \equiv 1.$$

Therefore

$$H_0(p) = c|p|.$$

□

The theorem is deliberately stated on $\mathbb{R}^3 \setminus \{0\}$. The function $|p|$ is not differentiable at $p = 0$, and the origin does not belong to the regular intrinsic null shell. This is consistent with the projective character of the boundary: the zero-staticity geometry retains momentum direction while the radial scale becomes homogeneous.

It is useful to isolate the role of the two boundary equations. The homogeneity equation

$$H_0 - p \cdot \nabla H_0 = 0$$

alone permits an arbitrary positive angular profile $f(n)$. Conversely, the constant-speed equation alone is an eikonal condition and does not by itself impose degree-one homogeneity. The Euclidean null rigidity arises from their simultaneous imposition.

For the relativistic massive branch

$$H_Q(p) = c \sqrt{|p|^2 + Q^2},$$

one has

$$H_Q - p \cdot \nabla H_Q = \frac{cQ^2}{\sqrt{|p|^2 + Q^2}}.$$

Hence

$$\frac{H_Q - p \cdot \nabla H_Q}{H_Q} = \frac{Q^2}{|p|^2 + Q^2}.$$

But the compact-state representation gives

$$S = \frac{Q}{\sqrt{|p|^2 + Q^2}},$$

so that

$$\frac{H_Q - p \cdot \nabla H_Q}{H_Q} = S^2.$$

Staticity therefore has a second dynamical interpretation:

$$S^2$$

is the normalized obstruction of the massive Hamiltonian to degree-one homogeneity.

In particular,

$$S \rightarrow 0$$

is equivalent, on the relativistic branch, to

$$\frac{H - p \cdot \nabla H}{H} \rightarrow 0.$$

The projective null boundary is thus simultaneously a homogeneity boundary of the massive dispersion relation.

3.2 Hessian degeneration and the relativistic branch

The same zero-staticity limit is visible at second order in momentum space. Let

$$R(p) := \sqrt{|p|^2 + Q^2},$$

so that

$$H_Q(p) = cR(p).$$

Its gradient is

$$\nabla H_Q = c \frac{p}{R},$$

and its Hessian is

$$\nabla^2 H_Q = \frac{c}{R} \left(I - \frac{pp^T}{R^2} \right).$$

At a point $p \neq 0$, decompose momentum space into the radial direction

$$\mathbb{R}p$$

and the two-dimensional transverse subspace

$$p^\perp.$$

For every transverse vector $w \in p^\perp$,

$$\nabla^2 H_Q w = \frac{c}{R} w.$$

Thus the transverse Hessian eigenvalue is

$$\lambda_{\perp} = \frac{c}{R},$$

with multiplicity two. In the radial direction,

$$\nabla^2 H_Q p = \frac{cQ^2}{R^3} p,$$

so that

$$\lambda_{\parallel} = \frac{cQ^2}{R^3}.$$

Their ratio is

$$\frac{\lambda_{\parallel}}{\lambda_{\perp}} = \frac{Q^2}{R^2}.$$

Using

$$S = \frac{Q}{R},$$

we obtain the exact identity

$$\frac{\lambda_{\parallel}}{\lambda_{\perp}} = S^2.$$

Proposition 3.2 (Staticity and Hessian anisotropy). *For the positive relativistic massive Hamiltonian*

$$H_Q(p) = c \sqrt{|p|^2 + Q^2},$$

the compact staticity satisfies

$$S^2 = \frac{H_Q - p \cdot \nabla H_Q}{H_Q} = \frac{\lambda_{\parallel}}{\lambda_{\perp}}.$$

Hence the same scalar measures both the normalized failure of degree-one homogeneity and the radial-to-transverse anisotropy of the Legendre differential.

The proposition gives a second-order interpretation of the zero-staticity boundary. As $S \rightarrow 0$,

$$\frac{\lambda_{\parallel}}{\lambda_{\perp}} \rightarrow 0.$$

The Legendre map therefore loses radial sensitivity relative to its transverse sensitivity.

For the intrinsic boundary Hamiltonian of Theorem 3.1,

$$H_0(p) = c|p|,$$

one has

$$\nabla^2 H_0 = \frac{c}{|p|} (I - nn^T), \quad n = \frac{p}{|p|}.$$

Consequently,

$$\nabla^2 H_0 p = 0,$$

while

$$\nabla^2 H_0 w = \frac{c}{|p|} w$$

for every $w \perp p$. The Hessian therefore has rank two and a one-dimensional radial kernel:

$$\ker \nabla^2 H_0 = \mathbb{R}p.$$

This kernel is not an accidental degeneracy. Differentiating the degree-one Euler identity

$$p \cdot \nabla H_0 = H_0$$

gives

$$\nabla^2 H_0 p = 0$$

for every C^2 degree-one homogeneous Hamiltonian. Thus the radial Hessian zero mode is the differential signature of projective homogeneity.

The massive and zero-staticity branches can therefore be compared as

$$\lambda_{\parallel} > 0, \quad S > 0,$$

in the massive interior, and

$$\lambda_{\parallel} = 0, \quad S = 0,$$

on the homogeneous boundary. The transverse response remains nonzero, while the radial response disappears.

Equivalently, the Legendre map

$$p \mapsto D = \frac{\nabla H}{c}$$

changes character at zero staticity. For $Q > 0$ it is locally nondegenerate on the relativistic massive branch. At $Q = 0$,

$$D(p) = \frac{p}{|p|}$$

depends only on direction, so that

$$D(\lambda p) = D(p), \quad \lambda > 0.$$

The radial momentum scale has become invisible to the compact motion coordinate. This is the dynamical counterpart of the projective identification of the null boundary.

In three momentum dimensions the determinant of the massive Hessian is

$$\det \nabla^2 H_Q = \lambda_{\parallel} \lambda_{\perp}^2 = \frac{c^3 Q^2}{R^5}.$$

Hence

$$\det \nabla^2 H_Q \rightarrow 0$$

as the zero-staticity limit is approached. The determinant measures the local volume distortion of the momentum-to-velocity Legendre map, while the eigenvalue ratio above identifies the specific radial direction in which the degeneration occurs.

The boundary can therefore be characterized dynamically by three equivalent features on the relativistic branch:

$$\begin{aligned} S &\rightarrow 0, \\ \frac{H - p \cdot \nabla H}{H} &\rightarrow 0, \end{aligned}$$

and

$$\frac{\lambda_{\parallel}}{\lambda_{\perp}} \rightarrow 0.$$

At the limiting intrinsic boundary Hamiltonian these become, respectively, zero staticity, exact degree-one homogeneity, and a radial Hessian kernel.

These conclusions are specific to the Euclidean compact-state branch. They do not imply that every compact convex motion–staticity body has the same massive interior merely because its zero-staticity boundary supports a homogeneous dispersion. In fact, the next section shows that a broad class of distinct compact convex bodies can possess the same projected boundary geometry and the same $Q = 0$ Hamiltonian. This boundary underdetermination is the reason additional response data will be required for the later boundary-to-bulk rigidity results.

4 Boundary Universality and Bulk Underdetermination

The Euclidean motion–staticity hemisphere is only one member of the compact convex class considered in the massive theory. Let

$$K \subset \mathbb{R}^3 \oplus \mathbb{R}$$

be a compact convex body and let

$$\Omega := \text{proj}_D K = \{D \in \mathbb{R}^3 : (D, S) \in K \text{ for some } S\}$$

denote its projection onto the motion sector. The upper boundary function is

$$\sigma_K(D) := \max \{S : (D, S) \in K\}, \quad D \in \Omega.$$

Convexity of K implies concavity of σ_K .

For the boundary interpretation considered here, we assume that the upper state shell is

$$\Sigma_K^+ = \{(D, \sigma_K(D)) : D \in \Omega\},$$

with

$$\sigma_K > 0 \quad \text{on} \quad \text{int } \Omega$$

and

$$\sigma_K = 0 \quad \text{on} \quad \partial\Omega.$$

The corresponding zero-staticity edge is therefore

$$\partial_0 \Sigma_K^+ = \{(D, 0) : D \in \partial\Omega\}.$$

For the Euclidean state body,

$$\Omega = \overline{B}^3, \quad \sigma(D) = \sqrt{1 - |D|^2},$$

and this edge reduces to the round sphere discussed in Section 2.

The compact-state convex duality theorem associates to K the Hamiltonian

$$H_K(p, Q) = c h_K(p, Q),$$

where

$$h_K(p, Q) = \sup_{(D,S) \in K} (p \cdot D + QS)$$

is the support function of K . Equivalently, for $Q > 0$,

$$H_K(p, Q) = c \sup_{D \in \Omega} [p \cdot D + Q\sigma_K(D)].$$

This formula exhibits separately the projected motion domain Ω and the vertical staticity profile σ_K . The distinction becomes decisive at $Q = 0$.

4.1 Projected convex bodies and boundary Hamiltonians

Let

$$h_\Omega(p) := \sup_{D \in \Omega} p \cdot D$$

denote the support function of the projected motion body Ω .

Theorem 4.1 (Projected-boundary Hamiltonian). *Let $K \subset \mathbb{R}^3 \oplus \mathbb{R}$ be compact and convex, and let $\Omega = \text{proj}_D K$. Then*

$$H_K(p, 0) = c h_\Omega(p).$$

Consequently, the $Q = 0$ Hamiltonian depends only on the projected motion body Ω and is independent of the vertical staticity profile of K .

Proof. By definition,

$$H_K(p, 0) = c \sup_{(D,S) \in K} p \cdot D.$$

The expression being maximized is independent of S . Hence the supremum over K reduces to the supremum over its projection:

$$\sup_{(D,S) \in K} p \cdot D = \sup_{D \in \Omega} p \cdot D = h_\Omega(p).$$

Therefore

$$H_K(p, 0) = c h_\Omega(p).$$

□

The theorem gives a precise formulation of the information that survives the zero-scale projection. The boundary Hamiltonian does not forget the entire compact geometry: because a closed convex body is uniquely determined by its support function, knowledge of

$$H_K(\cdot, 0)$$

for all momenta determines Ω uniquely. Explicitly,

$$\Omega = \bigcap_{p \in \mathbb{R}^3} \left\{ D : p \cdot D \leq \frac{H_K(p, 0)}{c} \right\}.$$

Thus the $Q = 0$ dispersion reconstructs the projected motion body.

What it does not reconstruct is the function

$$\sigma_K : \Omega \longrightarrow \mathbb{R},$$

which specifies how much staticity is assigned above each motion coordinate. For $Q > 0$, this information contributes directly:

$$H_K(p, Q) = c \sup_{D \in \Omega} [p \cdot D + Q \sigma_K(D)].$$

Two compact-state bodies may therefore possess the same Ω , and hence the same $Q = 0$ Hamiltonian, while producing different massive Hamiltonians for every positive value of Q .

Corollary 4.2 (Boundary equivalence). *Let K_1 and K_2 be compact convex state bodies with projected motion bodies Ω_1 and Ω_2 . Then*

$$H_{K_1}(p, 0) = H_{K_2}(p, 0) \quad \text{for all } p$$

if and only if

$$\Omega_1 = \Omega_2.$$

In particular, equality of the $Q = 0$ Hamiltonians does not imply

$$K_1 = K_2.$$

Proof. If $\Omega_1 = \Omega_2$, the conclusion follows immediately from Theorem 4.1. Conversely, equality of the $Q = 0$ Hamiltonians implies equality of the support functions

$$h_{\Omega_1} = h_{\Omega_2}.$$

A compact convex body is uniquely determined by its support function, and hence

$$\Omega_1 = \Omega_2.$$

The final statement follows because distinct vertical profiles over the same projected body can define distinct compact convex state bodies. $\square$

This motivates an equivalence relation on compact-state bodies:

$$K_1 \sim_{\partial} K_2$$

whenever

$$\text{proj}_D K_1 = \text{proj}_D K_2.$$

Equivalently,

$$K_1 \sim_{\partial} K_2 \iff H_{K_1}(p, 0) = H_{K_2}(p, 0) \quad \text{for all } p.$$

A boundary-equivalence class therefore consists of all massive compact-state geometries whose projected motion body is the same.

The distinction may be summarized as follows. The boundary support data determine

$$\Omega,$$

whereas the massive bulk additionally requires

$$\sigma_K.$$

The inverse map

$$H_K(\cdot, 0) \mapsto K$$

is therefore underdetermined, even though

$$H_K(\cdot, 0) \mapsto \Omega$$

is not.

This is the first boundary-to-bulk obstruction of the closed theory. A null or homogeneous boundary dispersion cannot by itself select a unique massive compact-state completion.

4.2 A non-Euclidean family with a common null boundary

The underdetermination can be exhibited explicitly using the non-Euclidean compact-state family already present in the convex duality construction. For

$$r > 1,$$

define

$$K_r = \{(D, S) \in \mathbb{R}^3 \oplus \mathbb{R} : |D|^r + |S|^r \leq 1\}.$$

Its projected motion body is independent of r :

$$\Omega_r = \{D : |D| \leq 1\} = \overline{B}^3.$$

The upper state shell is

$$S = \sigma_r(D) = (1 - |D|^r)^{1/r}, \quad |D| \leq 1.$$

Thus every member of the family has the same zero-staticity edge

$$|D| = 1, \quad S = 0,$$

even though its interior staticity profile depends on r .

Let

$$r' = \frac{r}{r-1}$$

be the Hölder-conjugate exponent. The support function of K_r is

$$h_{K_r}(p, Q) = (|p|^{r'} + |Q|^{r'})^{1/r'}.$$

For $Q > 0$ the corresponding positive Hamiltonian is therefore

$$H_r(p, Q) = c (|p|^{r'} + Q^{r'})^{1/r'}.$$

When

$$r = 2,$$

one has

$$r' = 2$$

and hence

$$H_2(p, Q) = c \sqrt{|p|^2 + Q^2},$$

which is the relativistic square-root Hamiltonian.

For a generic value

$$r \neq 2,$$

the massive dispersion is different:

$$H_r(p, Q) \neq c \sqrt{|p|^2 + Q^2}.$$

For example, at

$$|p| = Q > 0$$

one finds

$$H_r = cQ 2^{1/r'},$$

whose dependence on r is nontrivial. Hence the different members of the family define genuinely different massive Hamiltonians.

At $Q = 0$, however,

$$H_r(p, 0) = c \left(|p|^{r'} \right)^{1/r'} = c|p|$$

for every

$$r > 1.$$

Thus

$$H_r(p, 0) = c|p|$$

does not distinguish the Euclidean member $r = 2$ from any of its non-Euclidean relatives.

Proposition 4.3 (Common-boundary counterfamily). *The family*

$$K_r = \{(D, S) : |D|^r + |S|^r \leq 1\}, \quad r > 1,$$

has the following properties:

1. *every K_r has the same projected motion body*

$$\Omega_r = \overline{B}^3;$$

2. *every upper state shell has the same zero-staticity edge*

$$\partial_0 \Sigma_{K_r}^+ = \{(D, 0) : |D| = 1\};$$

3. *every member has the same $Q = 0$ Hamiltonian*

$$H_r(p, 0) = c|p|;$$

4. *for $Q > 0$, the Hamiltonians*

$$H_r(p, Q) = c \left(|p|^{r'} + Q^{r'} \right)^{1/r'}$$

are distinct for different values of r .

Consequently, a common projective null boundary and a common homogeneous boundary dispersion do not determine the massive bulk.

The Legendre representation makes the distinction visible directly in compact variables. Let

$$A = |p|^{r'} + Q^{r'}.$$

For $p \neq 0$,

$$D = \frac{\nabla_p H_r}{c} = |p|^{r'-2} p A^{1/r'-1},$$

and

$$S = \frac{H_r - p \cdot \nabla_p H_r}{cQ} = Q^{r'-1} A^{1/r'-1}.$$

Taking the r th powers gives

$$|D|^r = \frac{|p|^{r'}}{A},$$

and

$$S^r = \frac{Q^{r'}}{A}.$$

Therefore

$$|D|^r + S^r = 1,$$

as required.

For fixed $Q > 0$, the large-momentum limit satisfies

$$|D| \rightarrow 1, \quad S \rightarrow 0,$$

for every $r > 1$. Hence all members approach the same projected motion sphere. Nevertheless, the relation between the radial momentum scale, energy, and staticity remains r -dependent throughout the massive interior.

This example separates three statements that would otherwise be easy to conflate:

same boundary topology,

same boundary support function,

and

same massive dynamics.

The first two do not imply the third.

In particular, the Euclidean compact state is not characterized by the existence of the spherical zero-staticity boundary

$$|D| = 1$$

or by the limiting dispersion

$$H_0(p) = c|p|$$

alone. Both properties hold throughout the family K_r . A further condition must determine how the staticity coordinate extends away from the boundary and through the massive interior.

The next section introduces such a condition. Rather than postulating the Euclidean profile

$$S = \sqrt{1 - |D|^2},$$

we impose a scalar consistency relation between normalized energy and staticity. Together with regularity at the rest state, that relation will force the compact-state profile to be quadratic and thereby reduce the boundary-equivalence class to an ellipsoidal bulk.

5 Energy–Staticity Reciprocity and Quadric Rigidity

Section 4 showed that the projected zero-staticity boundary does not determine the massive compact-state interior. Distinct staticity profiles may have the same projected motion body and the same homogeneous $Q = 0$ Hamiltonian. Additional information is therefore required to distinguish the relativistic bulk from other compact convex completions.

The additional condition studied in this section is a scalar relation between normalized energy and staticity. It is not a general identity of Legendre mechanics and will be treated as an independent consistency condition. Its significance is that, together with regularity at the unique rest state, it forces the staticity profile to be quadratic and hence the compact state to be ellipsoidal.

Let

$$k := \frac{p}{Q}, \quad h(k) := \frac{H(p)}{cQ},$$

with $Q > 0$. On a regular Legendre branch,

$$D = \nabla_k h$$

and

$$S = h - k \cdot \nabla_k h.$$

Equivalently, if

$$u(D) := k \cdot D - h(k)$$

is the Legendre dual potential, then

$$u = -S, \quad k = \nabla_D u,$$

and

$$h = D \cdot \nabla_D u - u.$$

Since $u = -S$, the last identity becomes

$$h = S - D \cdot \nabla_D S.$$

For the standard relativistic branch,

$$h = \sqrt{1 + |k|^2}, \quad S = \frac{1}{\sqrt{1 + |k|^2}},$$

and therefore

$$hS = 1.$$

The following analysis asks what this relation implies before the Euclidean compact-state profile is assumed.

5.1 The reciprocity defect

Define the energy–staticity reciprocity defect by

$$\mathfrak{R}(D) := h(D)S(D) - 1.$$

Using

$$h = S - D \cdot \nabla S,$$

one obtains

$$\mathfrak{R} = S^2 - S D \cdot \nabla S - 1.$$

Introduce

$$\rho(D) := \frac{1 - S(D)^2}{2}$$

and let

$$E_D := D \cdot \nabla_D$$

denote the Euler vector field on motion space. Since

$$E_D \rho = -S D \cdot \nabla S,$$

the reciprocity defect satisfies the exact identity

$$\mathfrak{R} = (E_D - 2)\rho.$$

Proposition 5.1 (Reciprocity–homogeneity identity). *On every regular Legendre branch,*

$$\mathfrak{R} = hS - 1 = (D \cdot \nabla_D - 2) \frac{1 - S^2}{2}.$$

Consequently,

$$\mathfrak{R} = 0$$

if and only if

$$D \cdot \nabla_D(1 - S^2) = 2(1 - S^2).$$

Thus exact energy–staticity reciprocity is equivalent to degree-two homogeneity of $1 - S^2$ along the motion rays.

The proposition isolates the geometric content of the condition

$$hS = 1.$$

It does not initially impose a Euclidean norm on the motion coordinates. Instead, it requires

$$1 - S^2$$

to scale quadratically under

$$D \mapsto \lambda D.$$

Indeed, if Ω is convex and contains the rest state $D = 0$ in its interior, then every segment

$$\{tD : 0 \leq t \leq 1\}$$

lies in Ω . Along such a ray, exact reciprocity gives

$$\frac{d}{dt}\rho(tD) = \frac{2}{t}\rho(tD),$$

and hence

$$\rho(tD) = t^2\rho(D).$$

Equivalently,

$$1 - S(tD)^2 = t^2(1 - S(D)^2).$$

This already shows that the scalar reciprocity condition controls the entire radial dependence of the staticity profile. Its angular dependence, however, is not fixed by homogeneity alone. The latter will be constrained by regularity at the point at which all motion directions collapse to the unique rest state.

The defect also provides a useful measure of nonquadraticity. From

$$\mathfrak{R} = (E_D - 2)\rho,$$

if near the rest state

$$\rho = \rho_2 + \rho_3 + \rho_4 + \cdots,$$

where

$$E_D\rho_m = m\rho_m,$$

then

$$\mathfrak{R} = \rho_3 + 2\rho_4 + 3\rho_5 + \cdots.$$

The quadratic part ρ_2 is annihilated identically. Consequently, $\mathfrak{R}$ is insensitive to a change from one quadratic compact state to another, while it detects the genuinely nonquadratic components of the staticity profile.

This distinction will become useful later when affine recalibrations and nonquadratic boundary deformations are separated.

5.2 Rest regularity and ellipsoidal reconstruction

We now impose regularity at the unique rest state. Let

$$\Omega \subset \mathbb{R}^3$$

be a bounded convex domain containing the origin, and suppose

$$S \in C^2(\Omega) \cap C^0(\overline{\Omega})$$

satisfies

$$S > 0 \quad \text{in } \Omega,$$

$$S = 0 \quad \text{on } \partial\Omega,$$

and

$$S(0) = 1, \quad \nabla S(0) = 0.$$

The origin is thus the unique normalized rest state.

Theorem 5.2 (Energy–staticity quadric rigidity). *Under the preceding assumptions, suppose that the regular Legendre branch satisfies exact energy–staticity reciprocity,*

$$hS = 1$$

throughout Ω . Then there exists a positive-definite symmetric matrix A such that

$$S(D)^2 = 1 - D^T A D$$

and

$$\Omega = \{D \in \mathbb{R}^3 : D^T A D < 1\}.$$

Thus the compact motion–staticity state is an upper half of an ellipsoid.

Proof. Set

$$\rho(D) = \frac{1 - S(D)^2}{2}.$$

By Proposition 5.1, exact reciprocity implies

$$E_D \rho = 2\rho.$$

Hence

$$\rho(tD) = t^2 \rho(D)$$

whenever the radial segment from 0 to D lies in Ω . Convexity ensures this for every $D \in \Omega$ and $0 \leq t \leq 1$.

Since $S \in C^2$ at the rest state, ρ is also C^2 at $D = 0$. Moreover,

$$\rho(0) = 0, \quad \nabla \rho(0) = 0.$$

Taylor expansion at the origin gives

$$\rho(tD) = \frac{t^2}{2} D^T (D^2 \rho(0)) D + o(t^2).$$

On the other hand, degree-two homogeneity gives

$$\rho(tD) = t^2 \rho(D).$$

Dividing by t^2 and taking $t \rightarrow 0^+$ yields

$$\rho(D) = \frac{1}{2} D^T (D^2 \rho(0)) D.$$

Define

$$A := D^2 \rho(0).$$

Then

$$1 - S(D)^2 = D^T A D.$$

Because $S > 0$ in the interior,

$$D^T A D < 1$$

for $D \in \Omega$, while continuity and the boundary condition $S = 0$ imply

$$D^T A D = 1$$

for $D \in \partial\Omega$.

The domain Ω is bounded. Hence A cannot possess a nonzero null direction, since otherwise the level set

$$D^T A D < 1$$

would be unbounded along that direction. Therefore A is positive definite.

Finally, convexity and the presence of the origin imply that every ray meets $\partial\Omega$ exactly once. For a unit vector n , let this point be

$$r(n)n.$$

The boundary relation gives

$$r(n)^2 n^T A n = 1,$$

and hence

$$r(n) = \frac{1}{\sqrt{n^T A n}}.$$

These are precisely the radial endpoints of the ellipsoid

$$D^T A D < 1.$$

Thus

$$\Omega = \{D : D^T A D < 1\},$$

and

$$S(D) = \sqrt{1 - D^T A D}.$$

□

The matrix A is determined locally by the rest-state response. Since

$$\rho = \frac{1 - S^2}{2},$$

one has

$$A = D^2 \rho(0) = -D^2 S(0),$$

where the last equality uses

$$S(0) = 1, \quad \nabla S(0) = 0.$$

Thus the global ellipsoidal state is determined by the Hessian of the staticity profile at rest.

There is an equivalent Hamiltonian statement. Since

$$u(D) = -S(D) = -\sqrt{1 - D^T A D},$$

its gradient gives

$$k = \nabla_D u = \frac{AD}{S}.$$

Therefore

$$D = S A^{-1} k.$$

Substitution into

$$S^2 + D^T A D = 1$$

gives

$$S^2 (1 + k^T A^{-1} k) = 1.$$

Hence

$$S = \frac{1}{\sqrt{1 + k^T A^{-1} k}}.$$

Exact reciprocity,

$$hS = 1,$$

then yields

$$h(k) = \sqrt{1 + k^T A^{-1} k}.$$

Restoring physical units,

$$k = \frac{p}{Q}, \quad H = cQh,$$

one obtains

$$H(p) = c \sqrt{Q^2 + p^T A^{-1} p}.$$

Corollary 5.3 (Rest-response reconstruction). *Under the assumptions of Theorem 5.2, the entire compact-state profile and free Hamiltonian are determined by the quadratic response at the rest state. In particular,*

$$A = -D^2 S(0)$$

and

$$D_k^2 h(0) = A^{-1}.$$

The full dispersion relation is

$$h(k) = \sqrt{1 + k^T D_k^2 h(0) k}.$$

The theorem selects the class of ellipsoidal compact states, but it does not by itself select a preferred Euclidean representative. This is important: energy–staticity reciprocity fixes the quadratic character of the state geometry, not an independently specified spatial isotropy.

Suppose now that the motion coordinates transform in the ordinary vector representation of $SO(3)$ and that the compact-state geometry is rotationally invariant. Then

$$D^T A D$$

must satisfy

$$(RD)^T A (RD) = D^T A D$$

for every

$$R \in SO(3).$$

Equivalently,

$$R^T A R = A$$

for all rotations. Therefore

$$A = aI$$

for some positive constant a .

The motion coordinate is operationally calibrated by

$$D = \frac{v}{c}.$$

If the zero-staticity boundary corresponds to the universal limiting speed $|v| = c$, then

$$|D| = 1$$

on the boundary. Since

$$D^T A D = a|D|^2 = 1$$

there, one obtains

$$a = 1.$$

Corollary 5.4 (Round motion–staticity rigidity). *Assume the hypotheses of Theorem 5.2, together with ordinary spatial $SO(3)$ covariance and the calibration*

$$D = \frac{v}{c}$$

for which zero staticity occurs at $|v| = c$. Then

$$|D|^2 + S^2 = 1$$

and

$$H(p) = c \sqrt{|p|^2 + Q^2}.$$

The logical order is worth emphasizing. The round hemisphere has not been recovered from compactness alone. The successive inputs are instead

Legendre representation,

energy–staticity reciprocity,

C^2 regularity at the unique rest state,

and finally

$SO(3)$ covariance and speed calibration.

The first three select a quadric compact state; the last selects its round Euclidean representative.

This result provides a complementary rigidity route to the Hamiltonian compact-state consistency theorem of the massive theory. There the Euclidean compact state is specified first and Hamiltonian consistency selects the relativistic square root. Here one starts from a more general compact motion domain and a scalar energy–staticity relation; rest regularity first forces an ellipsoid, after which spatial calibration recovers the Euclidean state and the same relativistic Hamiltonian.

The preceding rigidity is deliberately stronger than the boundary information studied in Section 4. The common zero-staticity boundary of the family K_r does not determine its radial staticity profile, whereas the scalar condition

$$hS = 1$$

does. For the family

$$K_r = \{(D, S) : |D|^r + |S|^r \leq 1\},$$

one finds

$$hS = S^{2-r},$$

so exact reciprocity throughout the massive interior holds only for

$$r = 2.$$

Thus the reciprocity condition distinguishes the Euclidean member of a family whose projected null boundary and homogeneous $Q = 0$ dispersion are otherwise identical.

The next section weakens the quadric condition while retaining a controlled geometric completion. Instead of imposing exact energy–staticity reciprocity, we require the Legendre potential $u = -S$ to satisfy the hyperbolic affine-sphere volume equation. This produces a broader class of convex observer geometries and allows the reciprocity defect to be interpreted as a measure of their departure from the quadric case.

6 Affine-Sphere Completion and the Reciprocity Cubic

The rigidity result of Section 5 selects a quadric compact state once exact energy–staticity reciprocity and regularity at rest are imposed. It is useful, however, to place this quadric case inside a broader but still geometrically controlled class of convex observer states.

A natural such class is supplied by hyperbolic affine spheres. This construction is standard in affine differential geometry and has already been developed as a framework for Lorentz–Finsler observer geometry [?, ?]. In particular, vanishing mean Cartan torsion is equivalent to the observer indicatrix being a convex hyperbolic affine sphere centered at the origin, and the corresponding projective potential satisfies a Monge–Ampère equation.

The purpose of this section is not to rederive affine-sphere relativity. Instead, we ask what the motion–staticity scalar

$$\mathfrak{R} = hS - 1$$

measures once the staticity potential is placed inside this established affine-sphere class. The answer is that derivatives of $\mathfrak{R}$ directly reconstruct the cubic tensor measuring the departure of the observer indicatrix from a quadric.

6.1 Staticity as an affine-sphere potential

Let

$$\Omega \subset \mathbb{R}^n$$

be a bounded convex domain, with $n = 3$ in the physical motion space. Let

$$S : \Omega \longrightarrow (0, \infty)$$

be a smooth strictly concave staticity profile satisfying

$$S = 0 \quad \text{on} \quad \partial\Omega.$$

As in the Legendre formulation of Section 5, define

$$u(D) := -S(D).$$

Then

$$u < 0 \quad \text{in} \quad \Omega,$$

and u is strictly convex.

We now impose the additional affine-sphere condition

$$\det D^2 u = \rho^2 \left(-\frac{1}{u} \right)^{n+2},$$

where

$$\rho > 0$$

is a fixed affine-volume normalization.

Equation (6.1) is the standard Cheng–Yau Monge–Ampère equation for a hyperbolic affine sphere in projective coordinates. For a bounded convex domain Ω , the Cheng–Yau theorem gives, under the standard normalization, a unique convex solution satisfying

$$u = 0 \quad \text{on} \quad \partial\Omega.$$

The resulting affine sphere is asymptotic to the cone over Ω .

The corresponding centro-affine immersion is

$$F : \Omega \longrightarrow \mathbb{R}^{n+1}, \quad F(D) = -\frac{1}{u(D)}(1, D).$$

Since

$$u = -S,$$

this may equivalently be written as

$$F(D) = \frac{1}{S(D)}(1, D).$$

Thus the same projective map that sends the Euclidean motion–staticity hemisphere to the Lorentz hyperboloid extends, for a general convex staticity profile, to the standard projective representation of a centro-affine hypersurface.

For affine mean curvature normalized to

$$H_{\text{aff}} = -1,$$

the induced positive-definite affine metric is

$$g_{ij} = -\frac{u_{ij}}{u}.$$

In staticity variables,

$$g_{ij} = -\frac{S_{ij}}{S}.$$

The associated centro-affine connection has coefficients

$$\Gamma^k_{ij} = -\frac{1}{u} \left(u_i \delta_j^k + u_j \delta_i^k \right).$$

It is important to distinguish this metric from the compact Euclidean metric introduced in Section 2. The latter belongs to the specific round compact-state model. The metric g above is instead determined by the Hessian of the actual staticity potential and is defined for a general convex affine-sphere completion.

Define the centro-affine cubic tensor by

$$C_{ijk} := (\nabla_i g)_{jk}.$$

A direct calculation gives

$$C_{ijk} = -\frac{u_{ijk}}{u} - \frac{u_i u_{jk} + u_j u_{ik} + u_k u_{ij}}{u^2}.$$

The tensor is completely symmetric.

For later comparison with the conventional Pick normalization, we define

$$c_{ijk} := \frac{1}{2} C_{ijk}.$$

With this convention, c agrees with the Pick cubic used in the affine-sphere description of the indicatrix and, under the standard Lorentz–Finsler correspondence, with the pullback of the Cartan torsion to observer space [?].

When (6.1) holds, the Pick cubic is apolar:

$$g^{ij} c_{ijk} = 0.$$

The quadric case is characterized by

$$c_{ijk} = 0.$$

Thus c measures the genuinely nonquadratic part of the affine-sphere observer geometry.

The Euclidean motion–staticity state provides the elementary example. For

$$\Omega = B^n, \quad S(D) = \sqrt{1 - |D|^2},$$

one has

$$u(D) = -\sqrt{1 - |D|^2},$$

which satisfies the normalized equation

$$\det D^2 u = \left(-\frac{1}{u}\right)^{n+2}.$$

The corresponding affine sphere is the standard hyperboloid, and its Pick cubic vanishes identically.

6.2 Reciprocity defect and Pick-cubic tomography

We now return to the motion–staticity scalar

$$\mathfrak{R} = hS - 1.$$

Since

$$u = -S$$

and the Legendre relation gives

$$h = D \cdot \nabla u - u,$$

the reciprocity defect can be expressed entirely in terms of the affine potential:

$$\mathfrak{R} = u^2 - uD^k u_k - 1.$$

The following identity provides the central link between the scalar reciprocity response and the cubic affine geometry.

Theorem 6.1 (Reciprocity–cubic identity). *Let $u = -S$ be a smooth strictly convex Legendre potential on Ω , and let c_{ijk} be defined by (6.1) and (6.1). Then*

$$\partial_i \partial_j \mathfrak{R} = 2S^2 D^k c_{ijk}.$$

Equivalently, in terms of the unnormalized centro-affine cubic,

$$\partial_i \partial_j \mathfrak{R} = S^2 D^k C_{ijk}.$$

Proof. From

$$\mathfrak{R} = u^2 - uD^k u_k - 1,$$

one differentiation gives

$$\partial_i \mathfrak{R} = uu_i - u_i D^k u_k - u D^k u_{ki}.$$

Differentiating again and collecting terms yields

$$\partial_i \partial_j \mathfrak{R} = -D^k (uu_{ijk} + u_i u_{jk} + u_j u_{ik} + u_k u_{ij}).$$

Using (6.1),

$$u^2 C_{ijk} = -(uu_{ijk} + u_i u_{jk} + u_j u_{ik} + u_k u_{ij}),$$

and hence

$$\partial_i \partial_j \mathfrak{R} = u^2 D^k C_{ijk}.$$

Since

$$u^2 = S^2$$

and

$$C_{ijk} = 2c_{ijk},$$

the result follows. □

The theorem gives a tensorial interpretation to a quantity that was introduced purely as a scalar relation between energy and staticity. The Hessian of

$$hS - 1$$

is the radial contraction of the cubic tensor measuring nonquadratic observer anisotropy.

In particular, exact reciprocity

$$\mathfrak{R} = 0$$

implies

$$D^k c_{ijk} = 0$$

throughout the regular domain. This statement is weaker by itself than

$$c_{ijk} = 0,$$

but Theorem 5.2 shows that when exact reciprocity is combined with C^2 regularity at the unique rest state, the entire compact state is already forced to be quadric. In the affine-sphere subclass this is therefore consistent with

$$c_{ijk} = 0.$$

The identity becomes particularly transparent at the rest state. Choose affine observer coordinates such that

$$D = 0$$

is the normalized rest point and

$$u(0) = -1, \quad u_i(0) = 0.$$

Then

$$\mathfrak{R}(0) = 0, \quad \partial_i \mathfrak{R}(0) = 0,$$

and (6.1) gives

$$\partial_i \partial_j \mathfrak{R}(0) = 0.$$

Differentiating once more and evaluating at $D = 0$ yields

$$\partial_i \partial_j \partial_k \mathfrak{R}(0) = 2c_{ijk}(0).$$

Corollary 6.2 (Energy–staticity cubic tomography). *At a normalized rest state,*

$$c_{ijk}(0) = \frac{1}{2} \partial_i \partial_j \partial_k \mathfrak{R} \Big|_{D=0}.$$

Thus the local Pick cubic is reconstructed from the third velocity derivatives of the scalar energy–staticity reciprocity defect.

This formula supplies an operational interpretation of the affine cubic. Instead of introducing c_{ijk} as an independent tensorial observable, one may recover it from directional derivatives of the scalar quantity

$$\mathfrak{R} = \frac{HS}{cQ} - 1.$$

The statement is structural; it does not by itself assert that the required derivatives constitute a practical experimental protocol.

If the observer coordinates are further normalized so that

$$u_{ij}(0) = \delta_{ij},$$

then the local expansion takes the form

$$u(D) = -1 + \frac{1}{2}|D|^2 + \frac{1}{3}c_{ijk}(0)D^i D^j D^k + O(|D|^4).$$

Correspondingly,

$$\mathfrak{R}(D) = \frac{1}{3}c_{ijk}(0)D^i D^j D^k + O(|D|^4).$$

Thus quadratic modifications of the rest metric do not appear in the leading reciprocity defect. The first local contribution that detects genuinely nonquadratic anisotropy is cubic in the motion coordinate.

The exact identity also admits an integral form. Fix $D \in \Omega$ and consider the radial path

$$t \mapsto tD, \quad 0 \leq t \leq 1.$$

Let

$$\phi(t) = \mathfrak{R}(tD).$$

Using (6.1),

$$\phi''(t) = 2t S(tD)^2 c_{ijk}(tD) D^i D^j D^k.$$

Since

$$\phi(0) = 0, \quad \phi'(0) = 0,$$

integration gives

$$\mathfrak{R}(D) = 2 \int_0^1 t(1-t) S(tD)^2 c_{ijk}(tD) D^i D^j D^k dt.$$

Proposition 6.3 (Radial accumulation of cubic anisotropy). *The energy–staticity reciprocity defect at a state D equals the weighted radial accumulation of the Pick cubic between the rest state and D :*

$$\mathfrak{R}(D) = 2 \int_0^1 t(1-t) S(tD)^2 c_{tD}(D, D, D) dt.$$

This expression makes the distinction between local and integrated nonquadraticity explicit. The rest-state third jet measures the local cubic tensor, whereas the finite-state scalar $\mathfrak{R}(D)$ records a weighted accumulation of the same anisotropy along the corresponding motion ray.

A simple estimate follows immediately. With respect to any fixed affine-coordinate tensor norm,

$$|c_D(X, X, X)| \leq \|c(D)\| |X|^3,$$

one has

$$|\mathfrak{R}(D)| \leq 2|D|^3 \int_0^1 t(1-t) S(tD)^2 \|c(tD)\| dt.$$

Since

$$\int_0^1 t(1-t) dt = \frac{1}{6},$$

this implies

$$|\mathfrak{R}(D)| \leq \frac{|D|^3}{3} \sup_{0 \leq t \leq 1} [S(tD)^2 \|c(tD)\|].$$

Thus a finite departure from energy–staticity reciprocity requires a corresponding cubic anisotropy somewhere along the radial path from rest.

When the affine-sphere condition (6.1) is imposed, the Pick cubic is apolar. Contracting (6.1) with the inverse affine metric gives

$$g^{ij} \partial_i \partial_j \mathfrak{R} = 0.$$

Accordingly, on a fixed affine-sphere background the reciprocity defect satisfies a linear second-order elliptic equation whose coefficients are determined by the nonlinear staticity potential.

We define the corresponding affine Hessian operator by

$$\mathcal{L}_u f := g^{ij} \partial_i \partial_j f.$$

Then

$$\mathcal{L}_u \mathfrak{R} = 0.$$

This operator should not be confused with the Laplace–Beltrami operator of g , since the derivatives in its definition are taken in the distinguished affine coordinates. Its principal symbol is nevertheless positive definite, and the usual local elliptic maximum principle applies.

The elliptic relation will become important at the null boundary. If $\mathfrak{R}$ possesses suitable continuous boundary data, those data constrain its entire interior profile. In the next section we identify the corresponding renormalized boundary quantity and show that, under the affine-sphere completion, it is a direct reparametrization of the centro-affine curvature of $\partial\Omega$.

7 Renormalized Boundary Reciprocity

The preceding section identified the interior reciprocity defect

$$\mathfrak{R} = hS - 1$$

with the radial contraction of the Pick cubic. We now ask what information survives when the staticity tends to zero at the boundary of the projected motion domain.

The ordinary limit of h need not be finite. Indeed, on the relativistic massive branch,

$$h = \frac{1}{S},$$

so that $h \rightarrow \infty$ as $S \rightarrow 0^+$. The product

$$hS,$$

however, remains finite and equals unity in the quadric case. This motivates a renormalized boundary quantity that retains information lost by the projective boundary alone.

Let

$$\Omega \subset \mathbb{R}^n$$

be bounded, strictly convex, and contain the origin. Let

$$u = -S < 0$$

be an affine-sphere potential satisfying

$$\det D^2 u = \rho^2 \left(-\frac{1}{u} \right)^{n+2}, \quad u|_{\partial\Omega} = 0.$$

Throughout this section we assume sufficient smoothness and uniform convexity of $\partial\Omega$ for the boundary asymptotics stated below to hold.

For

$$\xi \in \partial\Omega,$$

let $\nu(\xi)$ denote the outward Euclidean unit normal, let $\kappa(\xi)$ denote the Euclidean Gauss curvature of $\partial\Omega$, and let

$$\delta(D) = \text{dist}(D, \partial\Omega)$$

denote the inward distance to the boundary in a tubular neighborhood of $\partial\Omega$.

The required asymptotic input is

$$S(D)^2 = 2\rho^{2/(n+1)}\kappa(\xi)^{-1/(n+1)}\delta(D) + o(\delta(D))$$

as $D \rightarrow \xi$ non-tangentially, together with the corresponding first-derivative asymptotic obtained by differentiating the leading term. This square-root behavior is characteristic of the singular Monge–Ampère boundary problem associated with hyperbolic affine spheres.

We define the renormalized boundary reciprocity response by

$$\beta(\xi) := \lim_{D \rightarrow \xi} \Re(D) = \lim_{D \rightarrow \xi} (h(D)S(D) - 1),$$

whenever the limit exists.

7.1 Boundary asymptotics and centro-affine curvature

We first derive the leading coefficient in the boundary asymptotic directly from the affine-sphere equation. Suppose locally that

$$S(D) \sim a(\xi) \sqrt{\delta(D)}.$$

Since

$$u = -S,$$

the normal second derivative has leading behavior

$$u_{\delta\delta} \sim \frac{a}{4\delta^{3/2}}.$$

In principal-curvature coordinates tangent to $\partial\Omega$, the tangential Hessian eigenvalues satisfy

$$u_{aa} \sim \frac{a\kappa_a}{2\sqrt{\delta}}, \quad a = 1, \dots, n-1,$$

where

$$\kappa = \prod_{a=1}^{n-1} \kappa_a.$$

Hence

$$\det D^2 u \sim \frac{a^n \kappa}{2^{n+1}} \delta^{-(n+2)/2}.$$

On the other hand,

$$\rho^2 \left(-\frac{1}{u} \right)^{n+2} \sim \rho^2 a^{-(n+2)} \delta^{-(n+2)/2}.$$

Matching the leading coefficients gives

$$a^{2n+2} = \frac{2^{n+1} \rho^2}{\kappa},$$

and therefore

$$a^2 = 2\rho^{2/(n+1)}\kappa^{-1/(n+1)}.$$

Thus

$$S^2 \sim 2\rho^{2/(n+1)}\kappa^{-1/(n+1)}\delta.$$

We now use the Legendre relation

$$h = S - D \cdot \nabla S.$$

It follows that

$$hS = S^2 - S D \cdot \nabla S = S^2 - \frac{1}{2} D \cdot \nabla(S^2).$$

Because the gradient of the inward distance satisfies

$$\nabla \delta \longrightarrow -\nu$$

at the boundary, (7.1) gives

$$S \nabla S \longrightarrow -\rho^{2/(n+1)}\kappa^{-1/(n+1)}\nu.$$

Since $S^2 \rightarrow 0$, one obtains

$$\lim_{D \rightarrow \xi} hS = \rho^{2/(n+1)}(\xi \cdot \nu)\kappa^{-1/(n+1)}.$$

Proposition 7.1 (Boundary reciprocity formula). *Under the preceding boundary regularity assumptions,*

$$1 + \beta(\xi) = \rho^{2/(n+1)}(\xi \cdot \nu(\xi))\kappa(\xi)^{-1/(n+1)}.$$

The right-hand side has a standard interpretation in centro-affine geometry. Let

$$s(\nu) := \sup_{D \in \Omega} D \cdot \nu$$

be the support function of Ω . At a smooth strictly convex boundary point with outward normal ν ,

$$s(\nu) = \xi \cdot \nu.$$

Define the centro-affine curvature by

$$\mathcal{K}_0(\nu) := \frac{\kappa(\nu)}{s(\nu)^{n+1}}.$$

Then Proposition 7.1 becomes

$$1 + \beta = \rho^{2/(n+1)}\mathcal{K}_0^{-1/(n+1)}.$$

Equivalently,

$$\mathcal{K}_0 = \frac{\rho^2}{(1 + \beta)^{n+1}}.$$

Theorem 7.2 (Boundary reciprocity–centro-affine identity). *Within the affine-sphere completion and under the stated boundary regularity assumptions, the renormalized motion–staticity boundary response is a reparametrization of the centro-affine curvature:*

$$\beta = \rho^{2/(n+1)}\mathcal{K}_0^{-1/(n+1)} - 1.$$

Thus β is determined entirely by the projective boundary shape and its second-order support geometry.

The theorem changes the interpretation of the boundary datum. The scalar β is not an additional arbitrary field placed on the null boundary. Once the affine-sphere condition is imposed, it is fixed by a classical centro-affine invariant of the projected motion body.

For the Euclidean unit ball,

$$s = 1, \quad \kappa = 1, \quad \rho = 1,$$

so that

$$\mathcal{K}_0 = 1$$

and

$$\beta = 0.$$

This agrees with exact relativistic reciprocity,

$$hS = 1.$$

More generally, a centered ellipsoid

$$\Omega = \{D : D^T A D < 1\}$$

with its compatible affine-sphere normalization also satisfies

$$hS = 1$$

throughout the interior. Hence

$$\beta = 0$$

on the boundary. The apparent dependence of (7.1) on the Euclidean curvature and support function is exactly compensated by the affine-volume normalization.

7.2 Boundary rigidity and support-response reconstruction

The centro-affine form of the boundary response immediately leads to a rigidity statement. A classical global theorem in centro-affine convex geometry states that a smooth closed strictly convex hypersurface enclosing the origin and having constant centro-affine curvature is a centered ellipsoid [?, ?].

Theorem 7.3 (Boundary reciprocity rigidity). *Assume the hypotheses of Theorem 7.2. If*

$$\beta$$

is constant on $\partial\Omega$, then

$$\mathcal{K}_0$$

is constant and Ω is a centered ellipsoid.

With the affine-sphere normalization compatible with

$$S(0) = 1,$$

the corresponding bulk is the quadric affine sphere and

$$\mathfrak{R} = 0$$

throughout Ω . In particular,

$$\beta = 0.$$

Proof. Equation (7.1) shows that constant β implies constant centro-affine curvature. The classical centro-affine rigidity theorem therefore implies that $\partial\Omega$ is a centered ellipsoid.

For an ellipsoidal domain, the normalized Cheng–Yau solution is quadratic in the sense that

$$S(D) = \sqrt{1 - D^T A D}$$

after the corresponding affine normalization. By Theorem 5.2, the associated Legendre Hamiltonian satisfies

$$hS = 1.$$

Therefore

$$\mathfrak{R} = 0$$

in the interior and

$$\beta = 0$$

at the boundary. □

There is a complementary interior proof in the special case

$$\beta = 0.$$

From Section 6, the affine-sphere reciprocity defect satisfies

$$\mathcal{L}_u \mathfrak{R} = 0.$$

If $\mathfrak{R}$ extends continuously to the boundary and

$$\mathfrak{R}|_{\partial\Omega} = 0,$$

the elliptic maximum principle gives

$$\mathfrak{R} = 0$$

throughout Ω . The quadric rigidity theorem of Section 5 then again yields an ellipsoid. Thus the same rigidity can be seen either from boundary centro-affine geometry or from the interior elliptic equation.

We next express the boundary datum directly in support coordinates. For a smooth strictly convex body,

$$D(\nu) = \nabla_{S^{n-1}} s(\nu) + s(\nu)\nu,$$

and its matrix of principal radii of curvature is

$$\nabla_{S^{n-1}}^2 s + s g_{S^{n-1}}.$$

Consequently,

$$\kappa^{-1} = \det_g \left(\nabla^2 s + s g \right),$$

where the determinant is taken relative to the round metric on S^{n-1} .

Substitution into (7.1) gives

$$1 + \beta(\nu) = \rho^{2/(n+1)} s(\nu) \det_g \left(\nabla^2 s + s g \right)^{1/(n+1)}.$$

Proposition 7.4 (Support-response representation). *The renormalized boundary reciprocity response can be reconstructed from the support function and its angular second derivatives:*

$$1 + \beta = \rho^{2/(n+1)} s \det_g \left(\nabla^2 s + s g \right)^{1/(n+1)}.$$

The support function itself has an immediate motion–staticity interpretation. By Theorem 4.1,

$$H_K(p, 0) = c h_\Omega(p).$$

For

$$p = \lambda v, \quad \lambda > 0,$$

positive homogeneity gives

$$h_\Omega(\lambda v) = \lambda s(v),$$

and hence

$$s(v) = \frac{H_K(\lambda v, 0)}{c \lambda}.$$

Thus the angular profile of the homogeneous boundary Hamiltonian determines the support function, while its second angular derivatives determine the centro-affine boundary response β .

This suggests a more general boundary-response primitive than the Lorentzian pairing with a preassigned round null vector. For an arbitrary convex projected motion body, define the linear motion response

$$R_v(D) = v \cdot D.$$

Its extremal response is

$$\sup_{D \in \Omega} R_v(D) = s(v).$$

The function s therefore records the complete convex boundary shape without presupposing a Minkowski metric or a round celestial sphere.

In the Euclidean Lorentzian case,

$$\Omega = B^n,$$

so that

$$s(v) = 1.$$

The more familiar null-direction description is recovered only after this round structure has been selected. The support response is therefore the appropriate convex generalization of the boundary directional response.

Solving (7.2) for the determinant gives

$$\det_g \left(\nabla^2 s + s g \right) = \rho^{-2} \left(\frac{1 + \beta}{s} \right)^{n+1}.$$

Hence prescribing β leads to a nonlinear Monge–Ampère equation for the boundary support function.

Proposition 7.5 (Boundary inverse problem). *Given a positive boundary function*

$$1 + \beta : S^{n-1} \longrightarrow (0, \infty),$$

reconstruction of a smooth strictly convex projected motion body within the affine-sphere class reduces to solving

$$\det_g(\nabla^2 s + s g) = \rho^{-2} \left(\frac{1 + \beta}{s} \right)^{n+1}.$$

In the standard L_p -Minkowski notation for a convex body in $\mathbb{R}^n$, this is the critical centro-affine case $p = -n$.

The proposition should not be interpreted as an unconditional global uniqueness theorem. Prescribed centro-affine curvature belongs to the critical L_p -Minkowski problem, whose existence and uniqueness properties depend on regularity, symmetry, normalization, and the prescribed data. In particular, nonuniqueness phenomena are known for general prescribed centro-affine curvature. The constant-curvature case used in Theorem 7.3 is substantially more rigid.

The correct boundary-to-bulk interpretation is therefore hierarchical. The homogeneous boundary Hamiltonian determines the support function

$$H(\cdot, 0) \longrightarrow s,$$

the support function determines the projected convex body

$$s \longrightarrow \Omega,$$

and, in the affine-sphere completion, the Cheng–Yau boundary-value problem determines the canonical staticity potential

$$\Omega \longrightarrow S.$$

The Legendre transform then reconstructs the corresponding massive Hamiltonian

$$S \longrightarrow h \longrightarrow H.$$

The boundary reciprocity response β provides an equivalent centro-affine description of the second-order boundary geometry and supplies a direct rigidity diagnostic for the quadric case.

The next section studies this reconstruction near the round Lorentzian state. There the nonlinear boundary equation linearizes to an explicit spherical operator, and its kernel can be identified exactly with the infinitesimal ellipsoidal recalibration modes.

8 Near-Quadric Boundary Tomography

The global boundary equation

$$1 + \beta = \rho^{2/(n+1)} s \det_g(\nabla^2 s + s g)^{1/(n+1)}$$

is nonlinear and, for general prescribed data, belongs to the critical centro-affine Minkowski problem. A considerably sharper statement is available near the quadric state.

In this section we linearize the boundary-response map around the round unit ball. The result identifies exactly which boundary deformations are visible to the renormalized reciprocity response. The only invisible infinitesimal modes are the degree-two spherical harmonics, which are precisely the volume-preserving ellipsoidal recalibrations of the round state.

We choose the affine normalization

$$\rho = 1$$

and take the reference projected motion body to be

$$\Omega_0 = B^n.$$

Its support function is

$$s_0 = 1,$$

its centro-affine curvature is

$$\mathcal{K}_0 = 1,$$

and its boundary reciprocity response vanishes:

$$\beta_0 = 0.$$

Consider a smooth one-parameter deformation

$$s_\varepsilon = 1 + \varepsilon f + O(\varepsilon^2),$$

where

$$f : S^{n-1} \longrightarrow \mathbb{R}.$$

The purpose of the linearized theory is to determine which components of f can be reconstructed from

$$\delta\beta := \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \beta_\varepsilon.$$

8.1 Linearized reciprocity response

Define

$$A[s] := \nabla^2 s + s g,$$

where g is the round metric on S^{n-1} and all covariant derivatives in this section are taken with respect to g . For the round state,

$$A[1] = g.$$

Under the perturbation

$$s_\varepsilon = 1 + \varepsilon f + O(\varepsilon^2),$$

one has

$$A[s_\varepsilon] = g + \varepsilon (\nabla^2 f + f g) + O(\varepsilon^2).$$

Using the standard first variation of the determinant,

$$\det(I + \varepsilon B) = 1 + \varepsilon \operatorname{tr} B + O(\varepsilon^2),$$

we obtain

$$\det_g A[s_\varepsilon] = 1 + \varepsilon (\Delta f + (n-1)f) + O(\varepsilon^2),$$

where

$$\Delta = g^{ab} \nabla_a \nabla_b$$

is the Laplace–Beltrami operator on S^{n-1} .

It follows that

$$\det_g A[s_\varepsilon]^{1/(n+1)} = 1 + \frac{\varepsilon}{n+1} (\Delta f + (n-1)f) + O(\varepsilon^2).$$

Multiplying by

$$s_\varepsilon = 1 + \varepsilon f + O(\varepsilon^2)$$

gives

$$1 + \beta_\varepsilon = 1 + \frac{\varepsilon}{n+1} (\Delta f + 2nf) + O(\varepsilon^2).$$

Theorem 8.1 (Linearized boundary reciprocity map). *At the round state, the Fréchet derivative of the support-to-reciprocity map is*

$$\delta\beta = \frac{1}{n+1} (\Delta_{S^{n-1}} + 2n) f.$$

Thus the linearized boundary response is governed by the shifted spherical Laplacian

$$\mathcal{B} := \frac{1}{n+1} (\Delta_{S^{n-1}} + 2n).$$

The same operator arises naturally from the linearization of centro-affine curvature. Since

$$\mathcal{K}_0 = (1 + \beta)^{-(n+1)}$$

in the normalization $\rho = 1$, one has

$$\delta\mathcal{K}_0 = -(n+1)\delta\beta.$$

Hence

$$\delta\mathcal{K}_0 = -(\Delta + 2n) f.$$

The shifted Laplacian therefore measures the first-order departure of the boundary from constant centro-affine curvature.

The support-function perturbation has a direct interpretation in terms of the homogeneous boundary Hamiltonian. From

$$H_0(\lambda\nu) = c\lambda s(\nu),$$

one obtains

$$H_{0,\varepsilon}(\lambda\nu) = c\lambda \left[1 + \varepsilon f(\nu) + O(\varepsilon^2) \right].$$

Thus f is the fractional angular perturbation of the zero-staticity dispersion, while

$$\delta\beta$$

is its second-order centro-affine response.

Theorem 8.1 therefore gives an explicit map

$$\text{boundary dispersion anisotropy} \longrightarrow \text{renormalized reciprocity anisotropy}.$$

To analyze this map spectrally, let

$$\mathcal{H}_\ell$$

denote the space of spherical harmonics of degree ℓ on S^{n-1} . With the sign convention used above,

$$\Delta Y_\ell = -\ell(\ell + n - 2)Y_\ell.$$

Hence $\mathcal{B}$ acts diagonally:

$$\mathcal{B}Y_\ell = \mu_\ell Y_\ell,$$

where

$$\mu_\ell = \frac{2n - \ell(\ell + n - 2)}{n + 1}.$$

The first few sectors have distinct geometric meanings. For $\ell = 0$,

$$\mu_0 = \frac{2n}{n + 1},$$

so an overall rescaling of the support function is visible when the affine-volume normalization ρ is held fixed.

For $\ell = 1$,

$$\mu_1 = 1.$$

These modes shift the convex body relative to the distinguished origin. They are visible because centro-affine geometry depends on the choice of center, which in the motion–staticity interpretation is fixed by the rest state.

For $\ell = 2$,

$$\mu_2 = 0.$$

This exceptional sector will be analyzed below.

For every

$$\ell \geq 3,$$

one has

$$\mu_\ell < 0.$$

Thus all higher nonquadratic angular deformations are detected by the linearized reciprocity response.

8.2 Ellipsoidal kernel and reconstruction

The kernel of the linearized boundary map is determined by

$$\ell(\ell + n - 2) = 2n.$$

Among nonnegative integers, the unique solution is

$$\ell = 2.$$

Therefore

$$\ker \mathcal{B} = \mathcal{H}_2.$$

This kernel has a direct geometric interpretation. Consider a one-parameter family of centered ellipsoids

$$\Omega_\varepsilon = \{D : D^T A_\varepsilon D < 1\},$$

where

$$A_\varepsilon = I + \varepsilon B + O(\varepsilon^2)$$

and B is symmetric. The support function is

$$s_\varepsilon(\nu) = \sqrt{\nu^T A_\varepsilon^{-1} \nu}.$$

Since

$$A_\varepsilon^{-1} = I - \varepsilon B + O(\varepsilon^2),$$

one finds

$$s_\varepsilon(\nu) = 1 - \frac{\varepsilon}{2} \nu^T B \nu + O(\varepsilon^2).$$

If the deformation preserves the affine volume to first order, then

$$\text{tr } B = 0.$$

The function

$$\nu \mapsto \nu^T B \nu$$

is then a degree-two spherical harmonic. Thus infinitesimal volume-preserving centered ellipsoids lie precisely in $\mathcal{H}_2$.

Conversely, every degree-two spherical harmonic can be represented as

$$f(\nu) = \nu^T B \nu$$

for a trace-free symmetric matrix B , up to an overall conventional normalization. Hence the full kernel of the linearized reciprocity map consists exactly of infinitesimal centered ellipsoidal recalibrations.

Theorem 8.2 (Ellipsoidal kernel). *For the linearized boundary-response operator*

$$\mathcal{B} = \frac{1}{n+1} (\Delta + 2n),$$

one has

$$\ker \mathcal{B} = \mathcal{H}_2.$$

This kernel is naturally identified with the infinitesimal volume-preserving linear deformations of the round ball into centered ellipsoids.

This result is consistent with the nonlinear rigidity established earlier. Exact energy–staticity reciprocity does not distinguish one ellipsoidal representative from another until a spatial calibration is chosen. At the linearized boundary level, the same freedom appears as the $\ell = 2$ kernel.

Let

$$f = \sum_{\ell=0}^{\infty} f_\ell$$

and

$$\delta\beta = \sum_{\ell=0}^{\infty} \beta_\ell,$$

where

$$f_\ell, \beta_\ell \in \mathcal{H}_\ell.$$

Equation (8.1) gives

$$\beta_\ell = \mu_\ell f_\ell.$$

Therefore, for every

$$\ell \neq 2,$$

the support deformation is reconstructed by

$$f_\ell = \frac{n+1}{2n-\ell(\ell+n-2)} \beta_\ell.$$

Corollary 8.3 (Linearized boundary tomography). *Modulo the degree-two ellipsoidal sector, the linearized renormalized reciprocity response uniquely determines the deformation of the boundary support function.*

More explicitly,

$$f = (n+1)(\Delta + 2n)^{-1} \delta\beta$$

on the orthogonal complement of

$$\mathcal{H}_2.$$

The reconstruction is quantitatively stable at the spectral level. For

$$\ell \neq 2,$$

one has

$$|\mu_\ell| \geq 1.$$

Indeed,

$$\mu_1 = 1,$$

while the $\ell = 0$ eigenvalue is larger than one and the absolute values of the eigenvalues for $\ell \geq 3$ increase with ℓ . Consequently, if

$$f \perp \mathcal{H}_2,$$

then

$$\|f\|_{L^2(S^{n-1})} \leq \|\delta\beta\|_{L^2(S^{n-1})}.$$

Higher-regularity versions follow from standard elliptic estimates for the shifted Laplacian.

Thus the failure of injectivity is finite-dimensional and geometrically controlled. There is no sequence of nonellipsoidal smooth modes becoming arbitrarily invisible at linear order.

For the physical case

$$n = 3,$$

the boundary is S^2 and

$$\delta\beta = \frac{1}{4} (\Delta_{S^2} + 6) f.$$

Writing

$$f(\theta, \phi) = \sum_{\ell, m} f_{\ell m} Y_{\ell m}(\theta, \phi),$$

one obtains

$$\delta\beta_{\ell m} = \frac{6 - \ell(\ell + 1)}{4} f_{\ell m}.$$

Hence

$$f_{\ell m} = \frac{4}{6 - \ell(\ell + 1)} \delta\beta_{\ell m}, \quad \ell \neq 2.$$

The exceptional sector is

$$\ell = 2,$$

for which

$$\delta\beta_{2m} = 0.$$

These five modes are precisely the five independent trace-free components of a symmetric 3×3 matrix and therefore match the infinitesimal $SL(3)$ deformation of the unit ball into a volume-preserving centered ellipsoid.

The mode interpretation may be summarized as follows:

$\ell = 0$: overall scale relative to the fixed affine normalization,

$\ell = 1$: displacement relative to the distinguished rest origin,

$\ell = 2$: ellipsoidal affine recalibration,

and

$\ell \geq 3$: genuine nonquadric boundary structure.

If additional central symmetry

$$\Omega = -\Omega$$

is imposed, all odd modes vanish and the first genuine nonellipsoidal sector occurs at

$$\ell = 4.$$

This decomposition gives a precise version of boundary tomography near the Lorentzian point. The projective null boundary by itself was shown in Section 4 to be insufficient to determine the massive interior. The renormalized reciprocity response adds second-order boundary information. At linear order, that information reconstructs every nonellipsoidal deformation of the support function while automatically quotienting the affine ellipsoidal freedom.

A full nonlinear local uniqueness theorem would require a gauge-fixed analysis of the centro-affine Minkowski equation

$$\det_g(\nabla^2 s + s g) = \left(\frac{1 + \beta}{s}\right)^{n+1}$$

modulo the relevant linear affine action. Such a theorem is not required for the present rigidity results and is not asserted here. The linearized analysis nevertheless establishes that the round state is nondegenerate in every direction transverse to the ellipsoidal orbit.

Together with the results of the preceding sections, the near-quadric theory yields the following boundary-to-bulk structure:

$$H(\cdot, 0) \longrightarrow s \longrightarrow \beta \longrightarrow \text{nonquadric boundary modes},$$

while the affine-sphere completion supplies

$$\Omega \longrightarrow S \longrightarrow H(\cdot, Q).$$

The final section collects these implications and distinguishes the universal projective boundary data from the additional response assumptions responsible for bulk rigidity.

9 Discussion and Boundary-to-Bulk Hierarchy

The closed motion–staticity construction developed in this paper was motivated by a simple question: what, if anything, does the zero-staticity boundary know about the massive interior?

The answer is hierarchical rather than binary. The projective boundary retains universal directional information, but this information alone is insufficient to reconstruct a massive compact-state geometry. Additional response data progressively reduce the allowed class of interiors. At the strongest level considered here, energy–staticity reciprocity and regularity at rest select a quadric state, while the affine-sphere completion turns the renormalized boundary reciprocity response into a centro-affine curvature observable.

The resulting structure does not modify the ordinary special-relativistic dispersion in the quadric branch. Its role is instead reconstructive: it separates which parts of the Lorentzian geometry follow from projective boundary structure, which require radial staticity information, and which depend on additional affine or isotropy assumptions.

9.1 What the null boundary determines

The weakest level of the construction is topological and projective. Closing the Euclidean compact state gives

$$\overline{S}_+ = \{(D, S) : |D|^2 + S^2 = 1, \quad S \geq 0\},$$

whose boundary

$$S = 0$$

is naturally identified with

$$S^2 \simeq \mathbb{P}(\mathcal{N}^+).$$

The staticity variable is simultaneously the compact-state scalar and a conformal defining function for the hyperbolic massive observer space.

At this level, the boundary records null direction but not the radial scale of a massive four-velocity. For fixed $Q > 0$,

$$S \rightarrow 0^+$$

corresponds to an infinite-rapidity limit of the massive branch. The finite object at the boundary is the projective null direction, not the diverging timelike representative.

A second level is supplied by the homogeneous boundary Hamiltonian. For a general compact convex state body K ,

$$H_K(p, Q) = c h_K(p, Q),$$

while

$$H_K(p, 0) = c h_\Omega(p), \quad \Omega = \text{proj}_D K.$$

Thus the complete $Q = 0$ dispersion determines the projected motion body Ω through its support function.

This boundary information is stronger than topology alone, but it still does not determine the vertical staticity profile

$$\sigma_K(D).$$

The family

$$K_r = \{(D, S) : |D|^r + |S|^r \leq 1\}$$

makes this failure explicit. Every member has the same projected motion ball and the same boundary Hamiltonian

$$H_r(p, 0) = c|p|,$$

while the massive dispersion

$$H_r(p, Q) = c(|p|^{r'} + Q^{r'})^{1/r'}$$

depends nontrivially on r .

The first general conclusion is therefore

$$\text{projective null boundary} \Rightarrow \text{unique massive bulk},$$

and even

$$\text{boundary support function} \Rightarrow \text{unique staticity profile}.$$

The missing information is radial. The energy–staticity reciprocity defect

$$\mathfrak{R} = hS - 1$$

supplies precisely such information. Its exact velocity-space form,

$$\mathfrak{R} = (D \cdot \nabla_D - 2) \frac{1 - S^2}{2},$$

shows that exact reciprocity is an Euler homogeneity condition on the staticity profile.

When

$$\mathfrak{R} = 0,$$

one has

$$1 - S^2$$

homogeneous of degree two. If all motion directions extend smoothly to a single C^2 rest state, this homogeneous function must be a quadratic form:

$$S^2 = 1 - D^T A D.$$

The massive bulk is therefore ellipsoidal.

This establishes a sharp distinction between boundary universality and bulk rigidity. The homogeneous boundary may be shared by many nonquadratic models, but the reciprocity condition constrains how the boundary is filled by massive states.

The affine-sphere completion introduces a further level of structure. Requiring

$$u = -S$$

to satisfy

$$\det D^2 u = \rho^2 \left(-\frac{1}{u} \right)^{n+2}$$

selects the canonical hyperbolic affine sphere associated with the projected convex domain. This equation and its interpretation in Lorentz–Finsler observer geometry belong to the established theory of hyperbolic affine spheres [?, ?].

Within this class, the motion–staticity reciprocity defect acquires a direct affine-geometric meaning. In the Pick normalization used here,

$$\partial_i \partial_j \mathfrak{R} = 2S^2 D^k c_{ijk}.$$

At the rest state,

$$c_{ijk}(0) = \frac{1}{2} \partial_i \partial_j \partial_k \mathfrak{R}(0).$$

Thus a scalar response quantity reconstructs the local cubic anisotropy of the observer indicatrix.

At the opposite end of the state space, the renormalized boundary value

$$\beta = \lim_{S \rightarrow 0} \mathfrak{R}$$

is related to the centro-affine curvature by

$$1 + \beta = \rho^{2/(n+1)} \mathcal{K}_0^{-1/(n+1)}.$$

Accordingly, the same reciprocity scalar connects the local rest geometry to the asymptotic boundary geometry.

The complete hierarchy considered in this paper may therefore be written schematically as

$$\begin{aligned} \text{boundary topology} &\longrightarrow \text{projective null directions,} \\ \text{boundary support response} &\longrightarrow \Omega, \\ \text{energy–staticity reciprocity} &\longrightarrow \text{radial bulk rigidity,} \\ \text{affine-sphere completion} &\longrightarrow \text{canonical convex bulk geometry,} \\ \text{renormalized boundary reciprocity} &\longrightarrow \text{centro-affine curvature.} \end{aligned}$$

None of the arrows should be interpreted as saying that the preceding level automatically implies the following one. Each level adds an explicit assumption or additional response datum. The value of the hierarchy is precisely that these inputs remain logically separated.

9.2 Rigidity, assumptions, and open problems

Several ingredients of the present construction are standard mathematical structures. The projective identification of the velocity ball with hyperbolic space, the future null sphere, the support-function representation of convex bodies, the Cheng–Yau hyperbolic affine-sphere equation, Pick cubic geometry, centro-affine curvature, and the centro-affine Minkowski problem all have established independent theories.

The contribution of the present construction is therefore not the introduction of these objects themselves. It is the particular response-based organization in which they arise from a compact motion–staticity state and its Legendre dynamics.

The main internal statements can be separated from the imported geometric theorems as follows. The projected-boundary identity

$$H_K(p, 0) = c h_\Omega(p)$$

follows directly from compact-state support duality. The reciprocity–homogeneity identity

$$hS - 1 = (D \cdot \nabla_D - 2) \frac{1 - S^2}{2}$$

is a consequence of the motion–staticity Legendre relations. Together with C^2 rest regularity, it gives the ellipsoidal rigidity theorem.

After the affine-sphere condition is added, the identity

$$\partial_i \partial_j \mathfrak{R} = 2S^2 D^k c_{ijk}$$

relates the same scalar defect to the standard Pick cubic. The boundary response

$$\beta$$

then reduces to a reparametrization of standard centro-affine curvature. Classical affine and convex rigidity theorems can subsequently be applied to the resulting boundary data.

This distinction is important for the interpretation of the results. Compactness alone does not derive Lorentz geometry. Nor does the existence of a spherical or projective null boundary select a unique massive dynamics. The quadric result requires the additional reciprocity condition

$$hS = 1$$

and regularity at rest. The further reduction from an arbitrary ellipsoid to the round state requires ordinary spatial $SO(3)$ covariance and the calibration of the limiting speed.

Similarly, the affine-sphere equation is not derived from compact-state duality. It is an additional volume-geometric completion. Once imposed, it provides a controlled class larger than the quadric sector and makes it possible to interpret departures from exact reciprocity in terms of established affine invariants.

The relation between the three principal geometric classes may be summarized as

$$\text{general compact convex states} \supset \text{affine-sphere states} \supset \text{quadric states} \supset \text{round Lorentz state}.$$

The inclusions correspond, respectively, to progressively stronger conditions on bulk volume geometry, cubic anisotropy, and spatial calibration.

The near-quadric analysis provides a local version of the same hierarchy. If

$$s = 1 + \varepsilon f$$

is a perturbation of the round support function, then

$$\delta\beta = \frac{1}{n+1} (\Delta_{S^{n-1}} + 2n) f.$$

The degree-two spherical harmonics form the complete kernel and represent infinitesimal centered ellipsoids. All modes transverse to this affine orbit are visible to the linearized boundary response.

Thus the boundary response does not attempt to distinguish geometries that are already equivalent at the chosen affine level. After fixing the corresponding affine calibration, the linearized map is invertible on the remaining nonquadric modes.

Several questions remain open.

First, the present boundary theory is classical. The projective null boundary should not be confused with a complete quantum massless-particle sector. In particular, the analysis does not address helicity representations, gauge reduction, polarization bundles, or quantum transport. These require additional internal state structure and are best treated separately.

Second, the limit

$$S \rightarrow 0$$

of a fixed- $Q > 0$ massive branch is a projective infinite-rapidity limit. It is not, by itself, a derivation of a finite-energy massless particle with $Q = 0$. The intrinsic zero-staticity Hamiltonian of Section 3 should therefore be understood as the dynamical structure compatible with the boundary equations, not as the literal endpoint of a fixed-energy massive trajectory.

Third, the general boundary inverse problem

$$\beta \mapsto s$$

is nonlinear and belongs to the critical centro-affine Minkowski problem. Global existence and uniqueness for arbitrary prescribed β are not asserted. The results established here are the constant-curvature rigidity theorem and the explicitly invertible near-round linearization modulo ellipsoidal modes.

Fourth, the boundary asymptotic formula used to identify β with centro-affine curvature requires the corresponding regularity of the Cheng–Yau solution near a smooth uniformly convex boundary. A fully self-contained treatment would require either a precise invocation of the strongest applicable boundary regularity theorem or a dedicated derivation of the required asymptotic with controlled remainder. This analytic issue is logically distinct from the algebraic reciprocity identities.

Fifth, no modified-relativity phenomenology is proposed here. The quadric and round branches reproduce ordinary relativistic kinematics. Nonquadric compact states may be studied as generalized convex or Lorentz–Finsler kinematics, but their physical viability would require independent analysis of symmetry, dynamics, causality, and experimental bounds.

The principal conclusion is therefore structural. The closed motion–staticity state admits a boundary-to-bulk organization in which different kinds of information enter at distinguishable levels:

directional boundary data, support data, radial reciprocity data, affine volume data.

Projective null geometry is the most universal part of this structure. Quadric Lorentz geometry is the most rigid.

In the round case these levels become mutually compatible:

$$S^2 = 1 - |D|^2,$$

$$hS = 1,$$

$$c_{ijk} = 0,$$

$$\beta = 0,$$

and

$$H(p) = c \sqrt{|p|^2 + Q^2}.$$

The Lorentzian state may therefore be characterized not only by its familiar hyperbolic geometry, but also as the common zero-defect point of bulk Legendre reciprocity, affine cubic anisotropy, and renormalized boundary curvature response.

This provides the main boundary-to-bulk interpretation of the closed motion–staticity framework.

10 Conclusion

This paper has developed a closed-boundary extension of compact motion–staticity geometry and used it to separate universal null-boundary structure from the additional information required to reconstruct a massive interior.

The first conclusion is negative but fundamental. For a general compact convex motion–staticity body K ,

$$H_K(p, Q) = c h_K(p, Q),$$

the zero-scale Hamiltonian is

$$H_K(p, 0) = c h_\Omega(p), \quad \Omega = \text{proj}_D K.$$

The $Q = 0$ dispersion therefore reconstructs the projected motion body Ω , but not the vertical staticity profile of K . Distinct massive theories may possess the same projective null boundary and the same homogeneous boundary Hamiltonian.

The explicit family

$$K_r = \{(D, S) : |D|^r + |S|^r \leq 1\}$$

demonstrates this underdetermination. Every member has the same projected unit ball and the same limiting Hamiltonian

$$H_r(p, 0) = c|p|,$$

while the massive dispersion

$$H_r(p, Q) = c(|p|^{r'} + Q^{r'})^{1/r'}$$

depends on r . Null-boundary universality is therefore weaker than massive bulk equivalence.

The additional scalar relation

$$hS = 1, \quad h = \frac{H}{cQ},$$

provides a rigidity mechanism. Its exact velocity-space form is

$$hS - 1 = (D \cdot \nabla_D - 2) \frac{1 - S^2}{2}.$$

Thus energy–staticity reciprocity is equivalent to degree-two homogeneity of $1 - S^2$. If all motion directions extend through a unique C^2 rest state, this homogeneous function must be quadratic:

$$S^2 = 1 - D^T A D.$$

The compact state is consequently ellipsoidal, and its full Hamiltonian is reconstructed as

$$H(p) = c \sqrt{Q^2 + p^T A^{-1} p}.$$

Ordinary spatial isotropy and the limiting-speed calibration reduce this family to

$$|D|^2 + S^2 = 1$$

and

$$H(p) = c \sqrt{|p|^2 + Q^2}.$$

A broader geometric completion is obtained by requiring

$$u = -S$$

to satisfy the hyperbolic affine-sphere Monge–Ampère equation. Within this established affine-sphere class, the motion–staticity reciprocity defect

$$\mathfrak{R} = hS - 1$$

has a direct tensorial interpretation:

$$\partial_i \partial_j \mathfrak{R} = 2S^2 D^k c_{ijk},$$

where c_{ijk} is the Pick cubic in the normalization adopted in this paper. At rest,

$$c_{ijk}(0) = \frac{1}{2} \partial_i \partial_j \partial_k \mathfrak{R}(0).$$

The nonquadratic anisotropy of the observer geometry can therefore be reconstructed from derivatives of a scalar energy–staticity response.

The same scalar possesses a boundary interpretation. Under the boundary regularity assumptions stated in Section 7, the renormalized response

$$\beta = \lim_{S \rightarrow 0} (hS - 1)$$

is related to the centro-affine curvature by

$$1 + \beta = \rho^{2/(n+1)} \mathcal{K}_0^{-1/(n+1)}.$$

Constant boundary reciprocity is therefore equivalent to constant centro-affine curvature and selects a centered ellipsoid under the standard global convexity hypotheses.

Near the round state, this nonlinear boundary relation becomes the explicit tomography operator

$$\delta\beta = \frac{1}{n+1} (\Delta_{S^{n-1}} + 2n) \delta s.$$

Its kernel consists precisely of degree-two spherical harmonics, which represent infinitesimal centered ellipsoidal recalibrations. Once these affine modes are quotiented or fixed, all remaining nonquadratic boundary modes are reconstructed uniquely at linear order.

The main outcome is therefore a boundary-to-bulk hierarchy:

$$\text{projective boundary} \longrightarrow \text{support geometry} \longrightarrow \text{reciprocity response} \longrightarrow \text{bulk rigidity}.$$

The successive arrows require successively stronger information. The projective null boundary is universal, while the Lorentzian quadric is rigid.

The framework does not claim that compactness alone derives special relativity, nor that the fixed- Q boundary constitutes a complete finite-energy massless particle theory. Hyperbolic affine spheres, centro-affine curvature, and Minkowski problems are also established mathematical structures rather than new objects introduced here. The distinctive content is their organization through a common motion–staticity and Legendre response architecture.

Within that architecture, the Lorentzian state is characterized simultaneously by

$$S^2 = 1 - |D|^2,$$

$$hS = 1,$$

$$c_{ijk} = 0,$$

$$\beta = 0,$$

and

$$H(p) = c \sqrt{|p|^2 + Q^2}.$$

These are not independent coincidences: they are different manifestations of the same quadric locus viewed respectively from compact-state geometry, Hamiltonian reciprocity, affine anisotropy, boundary curvature response, and momentum-space dynamics.

The analysis suggests two natural continuations. The first is a fully nonlinear stability theory for the centro-affine boundary inverse problem near the quadric orbit. The second is the extension of the closed motion–staticity boundary to internal spin degrees of freedom, where little-group contraction, helicity bundles, and massive-to-null transport become the relevant structures. These questions are logically subsequent to the classical boundary rigidity developed here and are left for separate work.

A Hamiltonian Form of Energy–Staticity Reciprocity

This appendix records the momentum-space form of the energy–staticity reciprocity condition and several consequences used implicitly in the main text.

Let

$$k = \frac{p}{Q}, \quad h(k) = \frac{H(p)}{cQ},$$

and assume that h is smooth on a Legendre-regular massive branch. The compact motion and staticity coordinates are

$$D = \nabla h$$

and

$$S = h - k \cdot \nabla h.$$

Define

$$\mathfrak{R} = hS - 1.$$

Introduce

$$q(k) = h(k)^2 - 1$$

and the momentum-space Euler vector field

$$E_k = k \cdot \nabla_k.$$

Then

$$E_k q = 2h E_k h.$$

Since

$$\mathfrak{R} = h(h - E_k h) - 1,$$

we obtain

$$\mathfrak{R} = q - \frac{1}{2}E_k q.$$

Equivalently,

$$\mathfrak{R} = -\frac{1}{2}(E_k - 2)q.$$

Proposition A.1 (Hamiltonian Euler form). *Exact energy–staticity reciprocity,*

$$hS = 1,$$

is equivalent to

$$E_k(h^2 - 1) = 2(h^2 - 1).$$

Thus

$$h^2 - 1$$

is homogeneous of degree two.

If

$$h(0) = 1, \quad \nabla h(0) = 0,$$

and h is C^2 at the rest momentum, then

$$q = h^2 - 1$$

is also C^2 there. Degree-two homogeneity gives

$$q(k) = \frac{1}{2}k^T D^2 q(0)k.$$

At rest,

$$D^2 q(0) = 2D^2 h(0),$$

and hence

$$q(k) = k^T G k, \quad G := D^2 h(0).$$

Therefore

$$h(k) = \sqrt{1 + k^T G k}.$$

Proposition A.2 (Momentum-space quadric rigidity). *Suppose*

$$hS = 1$$

on a connected Legendre-regular branch containing a normalized C^2 rest state

$$h(0) = 1, \quad \nabla h(0) = 0.$$

Then

$$h(k) = \sqrt{1 + k^T G k},$$

where

$$G = D^2 h(0).$$

If the Legendre map is locally nondegenerate at rest, then G is positive definite.

The corresponding velocity variables are

$$D = \nabla h = \frac{Gk}{h}$$

and, by reciprocity,

$$S = \frac{1}{h}.$$

Consequently,

$$D^T G^{-1} D = \frac{k^T G k}{h^2} = 1 - \frac{1}{h^2},$$

so that

$$D^T G^{-1} D + S^2 = 1.$$

This is the momentum-space counterpart of Theorem 5.2.

There is also a differential tensor form of reciprocity. Because

$$S = h - k \cdot \nabla h,$$

one has

$$\nabla S = -D^2 h k.$$

Differentiating

$$hS = 1$$

gives

$$S \nabla h + h \nabla S = 0.$$

Therefore

$$D^2 h k = \frac{S}{h} \nabla h.$$

Conversely, suppose

$$D^2 h k = \frac{S}{h} \nabla h.$$

Then

$$\nabla(hS) = S \nabla h - h D^2 h k = 0.$$

Thus hS is constant on each connected component. If the component contains the normalized rest state,

$$h(0) = S(0) = 1,$$

then

$$hS = 1.$$

Hence the scalar and tensor conditions are equivalent on the normalized connected branch:

$$hS = 1 \iff D^2 h k = \frac{S}{h} \nabla h.$$

For the quadric solution

$$h = \sqrt{1 + k^T G k},$$

the Hessian is

$$D^2h = \frac{G}{h} - \frac{Gkk^T G}{h^3}.$$

Relative to the rest metric G , it has one radial eigenvalue

$$\lambda_{\parallel} = S^3$$

and $n - 1$ transverse eigenvalues

$$\lambda_{\perp} = S.$$

More precisely, these are the eigenvalues of

$$G^{-1}D^2h$$

with respect to the G -orthogonal radial/transverse decomposition.

Thus

$$\frac{\lambda_{\parallel}}{\lambda_{\perp}} = S^2.$$

Furthermore, the matrix determinant lemma gives

$$\frac{\det D^2h(k)}{\det D^2h(0)} = S^{n+2}.$$

The scaled Hessian

$$hG^{-1}D^2h$$

has eigenvalues

$$S^2, 1, \dots, 1,$$

and therefore

$$\det(hG^{-1}D^2h) = S^2.$$

Thus, on the quadric branch, staticity may be recovered not only from the Legendre remainder but also from the normalized volume distortion of the Hamiltonian Legendre map.

Finally, the reciprocity defect admits an exact radial inversion. From (A), along a ray

$$t \mapsto tk$$

one has

$$\frac{d}{dt} \left[\frac{q(tk)}{t^2} \right] = -\frac{2\Re(tk)}{t^3}.$$

Assuming the rest Hessian exists,

$$\lim_{t \rightarrow 0} \frac{q(tk)}{t^2} = k^T G k.$$

Integration from 0 to 1 gives

$$h(k)^2 - 1 = k^T G k - 2 \int_0^1 \frac{\Re(tk)}{t^3} dt.$$

Hence the reciprocity defect reconstructs the nonquadratic part of the dispersion relation along each momentum ray.

For example, if

$$|\Re(tk)| \leq \varepsilon t^{2+\alpha} |k|^{2+\alpha}, \quad \alpha > 0,$$

then

$$|h(k)^2 - 1 - k^T G k| \leq \frac{2\varepsilon}{\alpha} |k|^{2+\alpha}.$$

This gives a simple quantitative form of near-quadric Hamiltonian stability.

B The Common-Boundary Family K_r

This appendix collects several formulas for the family

$$K_r = \{(D, S) \in \mathbb{R}^n \oplus \mathbb{R} : |D|^r + |S|^r \leq 1\}, \quad r > 1.$$

The family provides a useful test of which structures are truly fixed by the zero-staticity boundary.

Let

$$r' = \frac{r}{r-1}$$

be the Hölder-conjugate exponent. The support function of K_r is

$$h_{K_r}(p, Q) = (|p|^{r'} + |Q|^{r'})^{1/r'}.$$

For $Q > 0$ the corresponding Hamiltonian is

$$H_r(p, Q) = c (|p|^{r'} + Q^{r'})^{1/r'}.$$

In dimensionless variables

$$k = \frac{p}{Q}, \quad h_r(k) = \frac{H_r}{cQ},$$

one has

$$h_r(k) = (1 + |k|^{r'})^{1/r'}.$$

The motion coordinate is

$$D = \nabla h_r = |k|^{r'-2} k (1 + |k|^{r'})^{1/r'-1},$$

and the Legendre staticity is

$$S = h_r - k \cdot \nabla h_r = (1 + |k|^{r'})^{-1/r'}.$$

Consequently,

$$|D|^r = \frac{|k|^{r'}}{1 + |k|^{r'}},$$

while

$$S^r = \frac{1}{1 + |k|^{r'}}.$$

Thus

$$|D|^r + S^r = 1.$$

The upper staticity profile in velocity variables is therefore

$$S_r(D) = (1 - |D|^r)^{1/r}.$$

Although these profiles differ throughout the interior, their projected motion domains are identical:

$$\Omega_r = \overline{B}^n.$$

The homogeneous boundary Hamiltonian is likewise independent of r :

$$H_r(p, 0) = c|p|.$$

The normalized Legendre homogeneity defect is

$$\frac{h_r - k \cdot \nabla h_r}{h_r} = \frac{S}{h_r}.$$

Using (B) and (B), one finds

$$\frac{S}{h_r} = \frac{1}{1 + |k|^{r'}} = S^r.$$

Thus the Euclidean relativistic identity

$$\frac{h - k \cdot \nabla h}{h} = S^2$$

is recovered only for

$$r = 2.$$

The energy–staticity product is

$$h_r S = \left(1 + |k|^{r'}\right)^{(r-2)/r}.$$

Since

$$S = \left(1 + |k|^{r'}\right)^{-1/r},$$

this becomes

$$h_r S = S^{2-r}.$$

Hence the reciprocity defect is

$$\mathfrak{R}_r = S^{2-r} - 1.$$

It vanishes identically if and only if

$$r = 2.$$

The zero-staticity behavior is particularly discriminating. As

$$S \rightarrow 0^+,$$

one obtains

$$\lim_{S \rightarrow 0^+} \mathfrak{R}_r = -1, \quad 1 < r < 2,$$

while

$$\mathfrak{R}_2 = 0,$$

and

$$\mathfrak{R}_r \rightarrow +\infty, \quad r > 2.$$

Thus all K_r share the same projective boundary and the same boundary Hamiltonian, but the renormalized energy–staticity response separates the Euclidean member from the rest.

For a perturbation

$$r = 2 + \varepsilon,$$

one has

$$\mathfrak{R}_r = S^{-\varepsilon} - 1.$$

At fixed $0 < S < 1$,

$$\mathfrak{R}_r = -\varepsilon \log S + O(\varepsilon^2 \log^2 S).$$

The boundary therefore amplifies a small exponent deformation: even when $|\varepsilon|$ is small,

$$|\varepsilon \log S|$$

need not remain small as $S \rightarrow 0$.

The actual staticity Hessian metric

$$g_S = -\frac{1}{S} D^2 S$$

also distinguishes the family. Write

$$\rho = |D|.$$

For

$$S_r(\rho) = (1 - \rho^r)^{1/r},$$

the transverse eigenvalue of g_S is

$$g_{\perp}^{(r)} = \frac{\rho^{r-2}}{1 - \rho^r},$$

while the radial eigenvalue is

$$g_{\parallel}^{(r)} = \frac{(r-1)\rho^{r-2}}{(1 - \rho^r)^2}.$$

For the same projected unit ball, the Hilbert–Klein metric has eigenvalues

$$g_{H,\perp} = \frac{1}{1 - \rho^2},$$

and

$$g_{H,\parallel} = \frac{1}{(1 - \rho^2)^2}.$$

Therefore

$$\frac{g_{\perp}^{(r)}}{g_{H,\perp}} = \rho^{r-2} \frac{1 - \rho^2}{1 - \rho^r},$$

and

$$\frac{g_{\parallel}^{(r)}}{g_{H,\parallel}} = (r-1)\rho^{r-2} \left(\frac{1 - \rho^2}{1 - \rho^r} \right)^2.$$

At the null boundary,

$$\rho \rightarrow 1^-,$$

these ratios approach

$$\frac{g_{\perp}^{(r)}}{g_{H,\perp}} \longrightarrow \frac{2}{r},$$

and

$$\frac{g_{\parallel}^{(r)}}{g_{H,\parallel}} \longrightarrow \frac{4(r-1)}{r^2}.$$

Both limiting ratios equal unity simultaneously only for

$$r = 2.$$

At the rest state,

$$\rho \rightarrow 0,$$

the factor

$$\rho^{r-2}$$

shows that the staticity Hessian response is regular and nondegenerate only in the quadratic case. For $r > 2$ it becomes degenerate at rest, while for $1 < r < 2$ it becomes singular.

The family therefore separates several notions that coincide in the Lorentzian case:

common projective boundary,

common homogeneous boundary dispersion,

regular rest response,

energy–staticity reciprocity,

and

boundary–bulk metric consistency.

Only the first two are universal throughout the family.

C Spectral Structure of the Linearized Boundary Map

Consider the linearized boundary-response operator

$$\mathcal{B} = \frac{1}{n+1} (\Delta_{S^{n-1}} + 2n).$$

The spherical harmonic decomposition

$$L^2(S^{n-1}) = \widehat{\bigoplus_{\ell=0}^{\infty} \mathcal{H}_{\ell}}$$

diagonalizes $\mathcal{B}$.

For

$$Y_{\ell} \in \mathcal{H}_{\ell},$$

the Laplace–Beltrami eigenvalue equation is

$$\Delta Y_{\ell} = -\ell(\ell + n - 2)Y_{\ell}.$$

Hence

$$\mathcal{B}Y_\ell = \mu_\ell Y_\ell,$$

with

$$\mu_\ell = \frac{2n - \ell(\ell + n - 2)}{n + 1}.$$

The kernel equation

$$\mu_\ell = 0$$

is

$$\ell(\ell + n - 2) = 2n.$$

Its unique nonnegative integer solution is

$$\ell = 2.$$

Thus

$$\ker \mathcal{B} = \mathcal{H}_2.$$

The kernel agrees with infinitesimal centered ellipsoids. Let

$$A_\varepsilon = I + \varepsilon B + O(\varepsilon^2),$$

where B is symmetric and trace free. The ellipsoid

$$D^T A_\varepsilon D < 1$$

has support function

$$s_\varepsilon(\nu) = \sqrt{\nu^T A_\varepsilon^{-1} \nu}.$$

Since

$$A_\varepsilon^{-1} = I - \varepsilon B + O(\varepsilon^2),$$

one obtains

$$s_\varepsilon(\nu) = 1 - \frac{\varepsilon}{2} \nu^T B \nu + O(\varepsilon^2).$$

For

$$\text{tr } B = 0,$$

the function

$$\nu^T B \nu$$

is a degree-two spherical harmonic.

Conversely, every element of $\mathcal{H}_2$ is the restriction to the unit sphere of a trace-free homogeneous quadratic polynomial. Hence

$$\ker \mathcal{B}$$

is exactly the tangent space to the volume-preserving centered ellipsoid orbit through the ball.

Away from the kernel, there is a uniform spectral gap. One has

$$\mu_1 = 1,$$

and

$$\mu_0 = \frac{2n}{n+1} > 1.$$

For

$$\ell \geq 3,$$

$$|\mu_\ell| = \frac{\ell(\ell + n - 2) - 2n}{n + 1} \geq \frac{n + 3}{n + 1} > 1.$$

Therefore

$$|\mu_\ell| \geq 1$$

for every

$$\ell \neq 2.$$

Let

$$P_2$$

denote the L^2 -orthogonal projection onto $\mathcal{H}_2$. If

$$P_2 f = 0,$$

then

$$\|\mathcal{B}f\|_{L^2}^2 = \sum_{\ell \neq 2} |\mu_\ell|^2 \|f_\ell\|_{L^2}^2 \geq \sum_{\ell \neq 2} \|f_\ell\|_{L^2}^2.$$

Thus

$$\|f\|_{L^2} \leq \|\mathcal{B}f\|_{L^2}.$$

Since

$$\delta\beta = \mathcal{B}f,$$

this gives

$$\|f\|_{L^2} \leq \|\delta\beta\|_{L^2}$$

after fixing the ellipsoidal gauge.

The operator $\mathcal{B}$ is elliptic of order two. Standard elliptic estimates therefore give, for every $s \geq 0$,

$$\|f\|_{H^{s+2}} \leq C_s (\|\mathcal{B}f\|_{H^s} + \|f\|_{L^2}).$$

Using (C) on the orthogonal complement of $\mathcal{H}_2$ gives

$$\|f\|_{H^{s+2}} \leq C'_s \|\delta\beta\|_{H^s}.$$

Proposition C.1 (Linearized stability modulo ellipsoids). *After imposing*

$$P_2 f = 0,$$

the linearized boundary-response map

$$f \mapsto \delta\beta = \frac{1}{n+1}(\Delta + 2n)f$$

is an isomorphism between the corresponding Sobolev spaces, with the estimate

$$\|f\|_{H^{s+2}} \leq C_s \|\delta\beta\|_{H^s}.$$

In the physical case

$$n = 3,$$

one has

$$\mu_\ell = \frac{6 - \ell(\ell + 1)}{4}.$$

The first sectors are

$$\mu_0 = \frac{3}{2}, \quad \mu_1 = 1, \quad \mu_2 = 0, \quad \mu_3 = -\frac{3}{2}, \quad \mu_4 = -\frac{7}{2}.$$

Higher modes become increasingly visible to the differential boundary response.

If central symmetry

$$s(-\nu) = s(\nu)$$

is imposed, all odd harmonics disappear. After quotienting the $\ell = 2$ ellipsoidal sector, the first genuine deformation occurs at

$$\ell = 4.$$

For such centrally symmetric perturbations, the lowest nontrivial response eigenvalue has magnitude

$$\frac{7}{2}$$

in three dimensions.

The spectral analysis thus gives a quantitative form to the statement that the renormalized reciprocity response detects nonquadratic boundary structure. The sole first-order ambiguity is the finite-dimensional affine ellipsoid orbit.

D Boundary Asymptotics of the Affine-Sphere Potential

This appendix records the formal leading-order boundary calculation underlying Theorem 7.2. The calculation is valid whenever the Cheng–Yau potential admits the stated smooth square-root asymptotic near a uniformly convex boundary point. Establishing the sharpest regularity hypotheses for this expansion belongs to the analytic theory of singular Monge–Ampère equations and is not attempted here.

Let

$$\Omega \subset \mathbb{R}^n$$

have smooth strictly convex boundary, and let

$$u = -S$$

satisfy

$$\det D^2 u = \rho^2 \left(-\frac{1}{u} \right)^{n+2}, \quad u < 0, \quad u|_{\partial\Omega} = 0.$$

Fix

$$\xi \in \partial\Omega.$$

Let

$$\nu$$

be the outward Euclidean unit normal and let

$$\delta$$

denote inward distance to the boundary, so that

$$\nabla \delta = -\nu$$

at $\partial\Omega$.

Choose principal coordinates tangent to the boundary at ξ . Let

$$\kappa_1, \dots, \kappa_{n-1}$$

be the principal curvatures and

$$\kappa = \prod_{a=1}^{n-1} \kappa_a$$

the Euclidean Gauss curvature.

Assume the leading staticity behavior

$$S = a(\xi)\delta^{1/2} + o(\delta^{1/2})$$

with the corresponding differentiated expansion.

Since

$$u = -S,$$

the normal derivatives satisfy

$$u_\delta = -\frac{a}{2}\delta^{-1/2} + o(\delta^{-1/2}),$$

and

$$u_{\delta\delta} = \frac{a}{4}\delta^{-3/2} + o(\delta^{-3/2}).$$

For a principal tangent direction e_a , the signed-distance Hessian satisfies

$$D^2\delta(e_a, e_a) = -\kappa_a + O(\delta).$$

Hence

$$D^2u(e_a, e_a) = \frac{a\kappa_a}{2\delta^{1/2}} + o(\delta^{-1/2}).$$

Mixed normal–tangential terms contribute only at lower order to the determinant under the assumed regularity.

Therefore

$$\det D^2u = \frac{a^n \kappa}{2^{n+1}} \delta^{-(n+2)/2} (1 + o(1)).$$

On the right-hand side of (D),

$$\rho^2 \left(-\frac{1}{u} \right)^{n+2} = \rho^2 a^{-(n+2)} \delta^{-(n+2)/2} (1 + o(1)).$$

Matching the leading coefficients yields

$$\frac{a^n \kappa}{2^{n+1}} = \rho^2 a^{-(n+2)}.$$

Thus

$$a^{2n+2} = \frac{2^{n+1}\rho^2}{K},$$

and

$$a^2 = 2\rho^{2/(n+1)}K^{-1/(n+1)}.$$

Consequently,

$$S^2 = 2\rho^{2/(n+1)}K^{-1/(n+1)}\delta + o(\delta).$$

We now derive the boundary reciprocity limit. The Legendre relation gives

$$h = S - D \cdot \nabla S.$$

Therefore

$$hS = S^2 - \frac{1}{2}D \cdot \nabla(S^2).$$

From (D),

$$\nabla(S^2) = 2\rho^{2/(n+1)}K^{-1/(n+1)}\nabla\delta + o(1),$$

and hence

$$\nabla(S^2) \longrightarrow -2\rho^{2/(n+1)}K^{-1/(n+1)}\nu.$$

Taking

$$D \rightarrow \xi$$

in (D) gives

$$\lim_{D \rightarrow \xi} hS = \rho^{2/(n+1)}(\xi \cdot \nu)K^{-1/(n+1)}.$$

Writing

$$s(\nu) = \xi \cdot \nu$$

and

$$\mathcal{K}_0 = \frac{\kappa}{s^{n+1}},$$

one obtains

$$\lim_{D \rightarrow \xi} hS = \rho^{2/(n+1)}\mathcal{K}_0^{-1/(n+1)}.$$

Thus

$$1 + \beta = \rho^{2/(n+1)}\mathcal{K}_0^{-1/(n+1)}.$$

The calculation also shows why a renormalized product rather than h itself is natural at the boundary. From (D),

$$S = O(\delta^{1/2}),$$

whereas

$$|\nabla S| = O(\delta^{-1/2}).$$

Consequently,

$$h = S - D \cdot \nabla S = O(\delta^{-1/2}),$$

but

$$hS = O(1).$$

The divergence of normalized energy and the vanishing of staticity cancel to produce a finite boundary response.

For the unit ball,

$$\kappa = 1, \quad s = 1, \quad \rho = 1,$$

and

$$S^2 = 2\delta + O(\delta^2),$$

since

$$1 - |D|^2 = (1 - |D|)(1 + |D|) = 2\delta + O(\delta^2).$$

The general boundary coefficient (D) therefore reproduces the standard motion–staticity hemisphere.

The analysis in this appendix should be read as a conditional local derivation: once the square-root boundary expansion with controlled differentiated remainder is available, the centro-affine interpretation of the renormalized reciprocity response follows directly.